NED University of Engineering and Technology
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GPU Sharing Mechanisms for Multi-Tenant Inference Serving

Utilization, Latency Isolation and Engineering Trade-offs

Operating Systems CEP, Topic23
Systematic Evidence Synthesis and Engineering Trade-offs

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Bounded rapid systematic review of GPU sharing for multi-tenant inference
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Abstract

This report evaluates GPU sharing mechanisms for multi-tenant inference serving, focusing on utilization, latency isolation and engineering trade-offs. Searches across four academic databases and supplementary sources produced 4,189 occurrences and 4,075 canonical records after 114 duplicate links. Structured record screening excluded 3,493 records. A bounded source-assessment stage evaluated 23 report versions, excluded two training-focused reports and included 21 report versions representing 20 unique studies. The synthesis compares temporal scheduling, MPS-style concurrency, MIG partitioning, API interception/remoting and hybrid controls. It records 200 Q1–Q10 appraisal items, 52 source-linked evidence rows and two calculated quantities. Results support conditional mechanism selection based on workload, hardware, latency objective, interference tolerance and isolation boundary; no mechanism dominates every deployment condition.

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1 Background

GPU sharing is an operating-system and runtime decision about how concurrent clients use compute, memory and scheduling capacity on one physical accelerator. Throughput and model density can improve while request tails, interference, memory pressure or isolation worsen. The relevant comparison is therefore conditional on workload, hardware and baseline, rather than a single system-wide ranking.

The assigned families are temporal sharing, MPS-style concurrency, MIG partitions, API interception/remoting and hybrids. Product documentation supplies definitions, not measured inference gains. The reviewed primary reports supply the experimental observations.

Context and review methods: [1, 2, 3, 4, 5, 6].

2 Engineering problem and scope

Choose a sharing mechanism for concurrent inference on one physical GPU under specified latency, utilization, density and isolation requirements. Multi-GPU papers are eligible only when their local co-tenant inference evidence is separable. Mixed training/inference reports are assessed using inference results rather than relabelling training throughput. Remoting is relevant where the endpoint GPU is shared and tenant effects are evaluable.

Decision constraint	Implication
Tail-latency objective	A mean or isolated kernel latency cannot establish request P99/ P98 compliance.
Memory and model density	Additional functions may require model swapping or runtime sharing; transfer and trust costs must be retained.
Isolation boundary	Performance fairness, memory separation, fault containment and security are different claims.
Deployment environment	GPU generation, compiler, driver hooks, runtime changes and host CPU demand affect applicability.

3 Objectives and research questions

The objective is a traceable conditional recommendation that relates a mechanism to a workload and a sacrifice. The submitted protocol remains unchanged; later deviations are documented separately.

RQ	REVIEW QUESTIONS
RQ1	What time-slicing, MPS, MIG and API remoting mechanisms are evaluated for concurrent inference on one GPU, and how do sharing granularity, layer and isolation boundary classify them?
RQ2	How do studies measure utilization, throughput, density, average/tail latency, predictability and interference, and how comparable are their experimental methods?
RQ3	Under what GPU, workload, batch, tenant mix and configuration conditions are changes in utilization, throughput, density and latency reported?

RQ	REVIEW QUESTIONS
RQ4	What resource and fault isolation properties, trade-offs, implementation costs and conflicting results are reported, and what limits the evidence?
RQ5	Which approach or combination is supported for specified inference-serving conditions and priorities, with what confidence and alternatives?

4 Review methodology

The review examines operating-system and runtime mechanisms for concurrent inference on one physical GPU. Search exports retain their filenames, query descriptions and record-level provenance. Exact identifiers join occurrence, canonical record, report, study and outcome registers. A structured record-screening pass uses the available abstracts, titles, document types and documented screening rationales. Exclusions retain a criterion and evidence location; records outside the bounded source-assessment set are identified separately.

The comparative synthesis is a bounded rapid review of twenty included studies. It extends the earlier fifteen-report assessment with InferFair, ArkGPU, Dyna-P, KRISP and Tally, for which detailed evidence locations were available. These additions broaden fairness, dynamic partitioning and interference control. Selection of this comparative corpus does not imply that all other relevant literature was excluded on technical grounds. Records beyond the stopping rule remain separate from the synthesis corpus.

The final preparation used one consolidated screening, appraisal and extraction pass, supported by structured tools and source-linked consistency checks. Screening, quality appraisal and extraction were carried out as a single consolidated pass by one AI reviewer under team direction, followed by manual checking of the working files by team members Team members reviewed the report structure and supplied working files. Screening, appraisal and extraction were consolidated under one consistent decision rubric. The submitted protocol remains unchanged, and the rapid-review deviation is recorded as amendment B007. The stopping rule closes the twenty-study comparative analysis while retaining a transparent count of records that were not advanced to source assessment.

5 Current accounting and review status

Unit or stage	Current count
Imported occurrences	4,189
Applied duplicate links	114
Canonical records with a record-screening decision	4,075
Record-level exclusions in revised screening	3,493
Records not advanced under the bounded-review stopping rule	559
Source-assessed exclusion reports	2
Corpus report versions	21
Study IDs in the comparative corpus	20
Q1-Q10 item appraisals	200
Registered evidence rows	52

Unit or stage	Current count
Separately calculated quantities	2

The record-screening partition is $4,075 = 3,493+559+2+21$. This equation counts canonical records, rather than repeated exports or outcome rows. Twenty study IDs link 21 corpus report versions in the bounded synthesis. The linked journal extension is P023 and maps to S005. REEF conference and journal versions describe one underlying study and are linked rather than counted twice. The report-to-record map and the source version used accompany each study. The public identifiers are: P for report, S for unique study, R for screened record, E for evidence row and C for calculated quantity.

6 Protocol and documented deviations

Change	Implementation and consequence
Evidence integration	Imported search and assessment records retain source-level provenance. The present review applies one scope, outcome interpretation and Q1-Q10 rubric.
Search implementation	Executed expressions and fields remain distinct from planned ones. ACM and ScienceDirect historical count discrepancies remain in the search audit.
B007 rapid review amendment	The implemented method uses structured screening and a bounded twenty-study synthesis with a declared stopping rule.
Report versions	REEF conference and TOCS reports map to one study; the author identifies TOCS as an extension. ParvaGPU SC24 publication identity is verified, with manuscript-specific extraction identified.
Print presentation	Only supported extraction fields are printed. Only documented technical details are presented; unsupported values are not inferred.
Counting boundary	Records beyond the stopping rule are reported as not advanced to source assessment; they are not treated as technical exclusions or failed retrievals.

7 Academic databases and provenance

Source	Occurrences	Linked repeats	Canonical attribution
ACM	1582	30	1552
IEEE	102	9	93
Springer	587	3	584
ScienceDirect	1898	67	1831
Supplementary	20	5	15

ACM, IEEE Xplore, Springer and ScienceDirect are represented in the active imports. Supplementary sources are tracked separately. Cross-database duplicates are attributed to the surviving canonical record; per-source remaining counts are not independent studies.

Four represented platforms do not establish a reproducible complete four-database search, and 559 records passed the coarse screen and were not source-assessed.

8 Search strategy and executed query tables

The final search register contains the platform, exact expression, field scope, filters, recorded date and imported count used in the review. Dates are reported at the precision preserved in the search record. The workbook retains the complete run log and filenames.

Run	Exact query	Field and filters	Records used	Recorded date
IEEE-MPS	(GPU OR "graphics processing unit") AND (inference OR "model serving" OR "deep learning" OR DNN OR LLM) AND (MPS OR "multi-process service")	All metadata; 2020–2026; language/type checked during screening	25	11 Oct 2026
IEEE-MIG	(GPU OR "graphics processing unit") AND (inference OR "model serving" OR "deep learning" OR DNN OR LLM) AND (MIG OR "multi-instance GPU" OR "GPU partitioning")	All metadata; 2020–2026; language/type checked during screening	23	11 Oct 2026
IEEE-TEMP	(GPU OR "graphics processing unit") AND (inference OR "model serving" OR "deep learning" OR DNN OR LLM) AND ("time slicing" OR "time sharing" OR "context switching")	All metadata; 2020–2026; language/type checked during screening	45	11 Oct 2026
IEEE-API	(GPU OR "graphics processing unit") AND (inference OR "model serving" OR "deep learning" OR DNN OR LLM) AND ("API remoting" OR "CUDA remoting" OR "GPU proxy" OR "GPU virtualization")	All metadata; 2020–2026; language/type checked during screening	9	11 Oct 2026
SPR-MASTER	(GPU OR "graphics processing unit") AND (inference OR "model serving" OR "deep learning" OR DNN OR LLM) AND ("time slicing" OR "time sharing" OR "context switching" OR MPS OR "multi-process service" OR MIG OR "multi-instance GPU" OR "GPU partitioning" OR "API remoting" OR "CUDA remoting" OR "GPU proxy" OR "GPU virtualization")	General indexed search; English; research article/conference paper; 2020–2026	587	11 Oct 2026
ACM-MASTER	(GPU OR "graphics processing unit") AND (inference OR "model serving" OR "deep learning" OR DNN OR LLM) AND ("time slicing" OR "time sharing" OR "context switching" OR MPS OR "multi-process service" OR MIG OR "multi-instance GPU" OR "GPU partitioning" OR "API remoting" OR "CUDA remoting" OR "GPU proxy" OR "GPU virtualization")	All fields; 2020–2026	1,582	14 Oct 2026
SD-TEMP	(GPU OR "graphics processing unit") AND (inference OR "model serving" OR "deep learning") AND ("time slicing" OR "time sharing" OR "context switching")	Find articles with these terms; 2020–2026	1,053	14 Oct 2026
SD-MPS	(GPU OR "graphics processing unit") AND (inference OR "model serving" OR "deep learning") AND (MPS OR "multi-process service")	Find articles with these terms; 2020–2026	563	14 Oct 2026
SD-MIG	(GPU OR "graphics processing unit") AND (inference OR "model serving" OR "deep learning") AND (MIG OR "multi-instance GPU" OR "GPU partitioning")	Find articles with these terms; 2020–2026	241	14 Oct 2026
SD-API	(GPU OR "graphics processing unit") AND (inference OR "model serving" OR "deep learning") AND ("API remoting" OR "CUDA remoting" OR "GPU proxy" OR "GPU virtualization")	Find articles with these terms; 2020–2026	41	14 Oct 2026

The four IEEE runs contribute 102 occurrences, Springer 587, ACM 1,582 and the four ScienceDirect runs 1,898, for 4,169 database occurrences. Twenty supplementary records were logged separately. Displayed-hit notes and overlapping export rows are retained in the workbook, while the counts above are the occurrences actually used in the consolidated import.

9 Literature period

The main period is 2020 through the actual 2026 search date for each run. Search logs record 11 October 2026 for IEEE and Springer and 14 October 2026 for ACM and ScienceDirect. Those dates are not treated as one universal cutoff. Later primary-paper retrieval is source assessment, not a claim that an earlier database run covered later publications. Older seminal exceptions require an individually documented justification; they are not silently mixed into the main window.

10 Inclusion and exclusion criteria

IC1-IC6 are conjunctive. At assessed full text, one decisive exclusion reason is recorded in protocol order. EC6 is applied only after a report has been assessed against the required technical detail.

IC	Required condition
IC1	English peer-reviewed full primary conference/journal report; final source identity checked.
IC2	Evaluated sharing among concurrent tenants/models/processes on one physical GPU, including temporal/hybrid sharing.
IC3	Inference/model serving; mixed workloads need separable inference outcomes.
IC4	Relevant inspectable outcome with interpretable experimental configuration.
IC5	Extractable within-one-GPU evidence and sufficient source detail; multi-GPU scale alone does not exclude.
IC6	Main publication window or a justified older seminal exception.

EC	Decisive exclusion condition
EC1	Non-primary/non-peer-reviewed version, survey, tutorial, editorial, preprint-only or abstract/poster item.
EC2	Training-only or no evaluable concurrent inference-sharing mechanism.
EC3	Hardware/microarchitecture-only work without evaluated systems-level sharing.
EC4	Only aggregate cluster/multi-GPU/remote-execution results without extractable local co-tenant evidence.
EC5	Non-GPU accelerator without a relevant evaluated GPU mechanism.
EC6	Assessed full text has insufficient methods/configuration/outcomes for any RQ.
EC7	Outside period without accepted exception or non-English full text.

11 PRISMA 2020 study-selection accounting

Figure 1 gives reconciled study-selection accounting for the bounded rapid review. The lower branch states the twenty-study synthesis corpus and records the stopping-rule branch separately. This makes the completed review boundary explicit without treating records beyond that boundary as technical exclusions or failed retrievals.

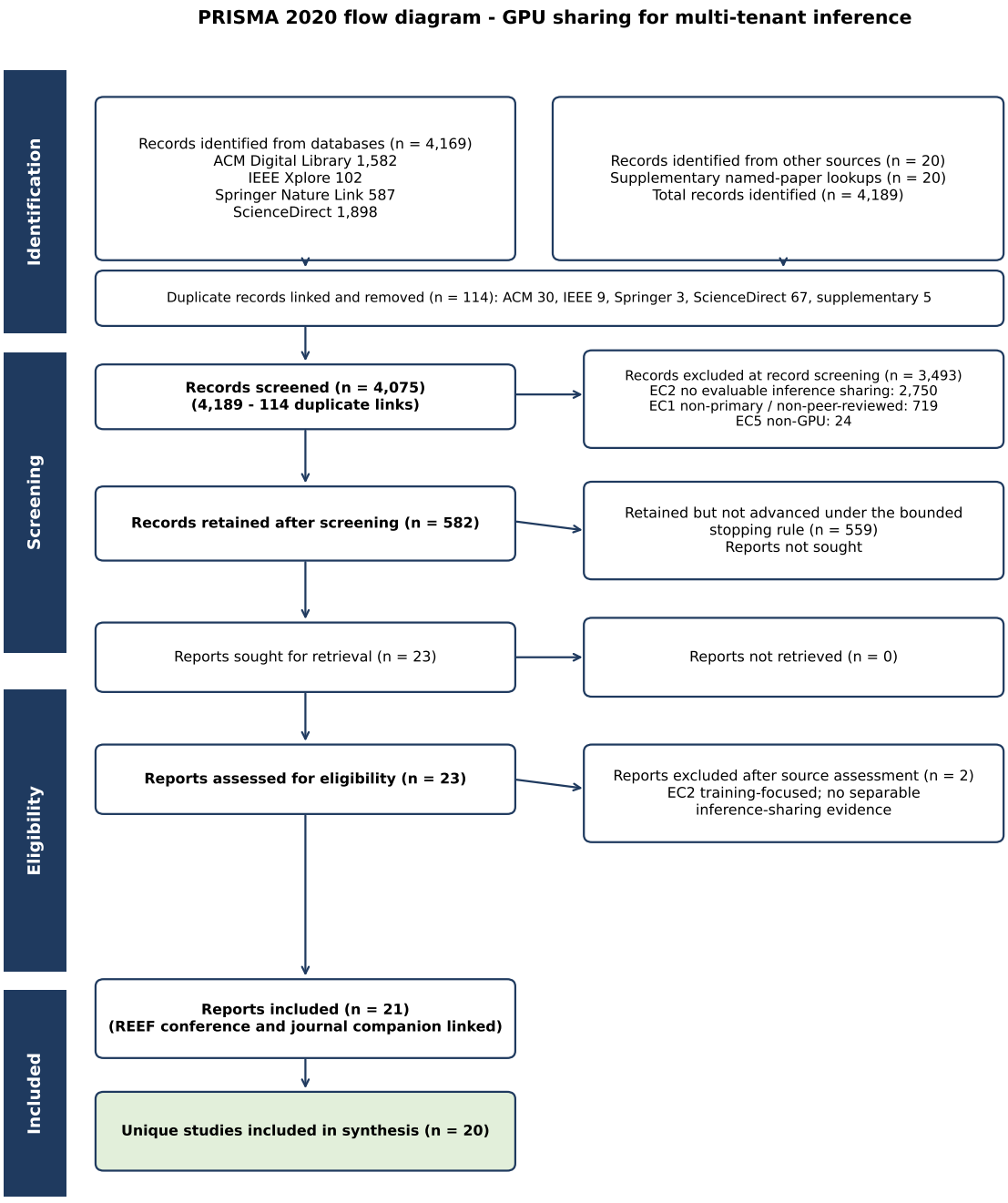


Figure 1: PRISMA-style study-selection accounting for the bounded review. The stopping-rule branch is reported separately from technical exclusions and unsuccessful retrievals.

Source-assessed exclusion	Decisive evidence
P016 TGS	Training-focused GPU container sharing; evaluation concerns training jobs. Evidence locations: printed pp. 69, 75-81; PDF pp. 2, 8-14.
P017 HiveD	Cluster reservation and affinity evaluated with training queue/completion-time behavior; no separable local concurrent inference outcome. Printed pp. 515, 521-525; PDF pp. 2, 8-12.

These two source decisions are distinct from metadata-level exclusions [7, 8]. These two source decisions are distinct from metadata-level exclusions [7, 8].

The twenty included studies are the studies that met the criteria within the bounded source-assessment set (23 reports assessed). The count is not a cap: 559 records retained at screening were not source-assessed and remain a limitation, as does the single-reviewer design.

12 Search, screening, duplicate and abstract logs

The accompanying workbook and CSV files preserve the 4,075 canonical records, all 4,189 occurrence-to-canonical links, and the source of each revised screening decision. The revised pass recovered abstracts for 1,142 of the 4,075 canonical records from original exports. The remaining 2,933 records (about 72%) had no abstract and were screened on title and document type only. Exact normalized DOI/title links connect 3,860 canonical records to the project evidence register. This is provenance coverage; each full-text assessment remains separately identified. Explicit document-type and abstract-scope rules complement those source-linked judgments.

Screening decisions are kept separate from retrieval and source eligibility. The revised export gives the decision, IC/EC basis, source path, decisions were generated in one batch passed from 13 Oct 2026 to 16 Oct 2026 using structured tools and associated report ID. It preserves the prior logs as an unchanged baseline. The complete electronic logs are supplied as data files rather than printing thousands of repeated rows in the main report.

13 Quality assessment

The review uses the submitted ten-item rubric with scores 0, 1 and 2. A justified claim-specific N/A is removed from the applicable denominator; it is not scored as zero. The raw sum, applicable maximum and normalized percentage are retained. Evidence-specific limits affect the relevant item score and interpretation. Scores recorded on other scales are not converted into the submitted Q1–Q10 rubric by multiplication.

Report	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10	Total	Norm.
P001	2	2	1	1	1	2	NA	2	1	1	13/18	72.2%
P002	2	2	2	1	1	2	2	2	1	2	17/20	85.0%
P003	2	2	2	2	2	2	2	2	1	1	18/20	90.0%
P004	2	2	2	1	2	2	2	2	1	1	17/20	85.0%
P005	2	2	1	2	2	2	1	2	1	2	17/20	85.0%
P006	2	2	1	2	1	2	2	2	NA	2	16/18	88.9%

Report	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10	Total	Norm.
P007	2	2	2	2	1	2	2	2	2	2	19/20	95.0%
P008	2	2	1	2	2	2	1	2	2	1	17/20	85.0%
P009	2	2	1	1	1	2	2	2	1	2	16/20	80.0%
P010	2	2	1	2	1	2	2	2	1	2	17/20	85.0%
P011	2	2	1	1	1	2	1	2	1	1	14/20	70.0%
P012	2	2	1	1	1	2	2	2	1	1	15/20	75.0%
P013	2	2	1	1	1	2	1	2	1	2	15/20	75.0%
P014	2	2	1	2	1	2	2	2	1	1	16/20	80.0%
P015	2	2	1	1	2	2	2	2	NA	2	16/18	88.9%
P018	2	2	1	2	1	2	1	2	1	1	15/20	75.0%
P019	2	2	1	2	1	2	2	2	1	1	16/20	80.0%
P020	2	2	1	2	1	2	1	2	1	1	15/20	75.0%
P021	2	2	1	2	1	2	2	2	1	1	16/20	80.0%
P022	2	2	1	2	1	2	2	2	1	2	17/20	85.0%

All twenty source-assessed studies score at least 60 percent of applicable points under the current rubric. Removing studies below that prespecified sensitivity threshold therefore leaves this corpus unchanged. This check does not remove the single-assessor, source-version or incomplete-search limitations. Item-level reasons and anchors follow in Appendix A.

14 Topic-specific extraction and study identities

The twenty-study corpus covers temporal preemption, MPS-based sharing, MIG-related designs, API interception/remoting and hybrid memory/runtime approaches. The extraction distinguishes physical GPU conditions, concurrent workloads, baseline, metric, unit, performance/resource isolation, overhead and limitations. The paper cards print supported fields and explicit interpretation boundaries. The original full extraction schema remains in the audit data.

A source using several GPUs qualifies only through identifiable local sharing evidence; aggregate cluster throughput is labelled separately. Mixed training/inference studies are interpreted using the separable inference outcome. Mean latency, p95/p99, job completion, goodput, aggregate throughput and memory savings remain separate metrics. The corpus register is:

Study	Report	Primary source	Year
S001	P001	VastPipe	2025
S002	P002	iGniter	2023
S003	P003	gpulets	2022
S004	P004	PipeSwitch	2020
S005	P005	REEF conference	2022
S006	P006	Paella: Low-latency Model Serving with Software-defined GPU Scheduling	2023
S007	P007	Orion: Interference-aware, Fine-grained GPU Sharing for ML Applications	2024
S008	P008	GSLICE: Controlled Spatial Sharing of GPUs for a Scalable Inference Platform	2020

Study	Report	Primary source	Year
S009	P009	Serving DNNs like Clockwork: Performance Predictability from the Bottom Up	2020
S010	P010	Colocating ML Inference and Training with Fast GPU Memory Handover (SIRIUS)	2025
S011	P011	ParvaGPU: Efficient Spatial GPU Sharing for Large-Scale DNN Inference in Cloud Environments	2024
S012	P012	Torpor: GPU-Enabled Serverless Computing for Low-Latency, Resource-Efficient Inference	2025
S013	P013	StreamBox: A Lightweight GPU SandBox for Serverless Inference Workflow	2024
S014	P014	Efficient Performance-Aware GPU Sharing with Compatibility and Isolation through Kernel Space Interception (KRYPTON)	2025
S015	P015	XSched: Preemptive Scheduling for Diverse XPUs	2025
S016	P018	InferFair: Towards QoS-aware scheduling for performance isolation guarantee in heterogeneous model serving systems	2024
S017	P019	ArkGPU: enabling applications' high-goodput co-location execution on multitasking GPUs	2023
S018	P020	Dyna-P: placement-aware dynamic partitioning for lightweight applications with modern GPUs	2025
S019	P021	KRISP: Enabling Kernel-wise RIght-sizing for Spatial Partitioned GPU Inference Servers	2023
S020	P022	Tally: Non-Intrusive Performance Isolation for Concurrent Deep Learning Workloads	2025

REEF conference evidence remains anchored to OSDI 2022; the TOCS extension is linked to the same S005 identity . ParvaGPU is a verified SC24 publication; the detailed technical extraction uses its author manuscript, so source-version equivalence is a limitation rather than an invented final-PDF check.

15 Taxonomy and integration layers

Family	Control boundary and evidence
Temporal and preemptive	PipeSwitch, REEF, Clockwork, Paella, Orion and XSched use different switch, admission or priority controls. Their kernel, request and model boundaries differ.
MPS and software spatial control	VastPipe, iGniter, gpulets and GSLICE allocate or regulate concurrent work. KRISP provides an AMD right-sizing comparison rather than NVIDIA MPS evidence.
MIG and hybrid partitioning	ParvaGPU combines MIG and MPS; KRYPTON integrates partitions with time control; Dyna-P studies placement and reconfiguration.
API interception and remoting	Torpor, StreamBox, KRYPTON, Tally and ArkGPU intervene at different runtime/driver boundaries, with different compatibility and trust assumptions.
Memory and runtime hybrids	SIRIUS hands memory between inference and training; StreamBox shares runtime resources within workflows.

Family	Control boundary and evidence
SLO fairness control	InferFair balances per-model slowdown and goodput; its fairness metric is not a p99 or hostile-isolation metric.

A family describes the control mechanism, not a guarantee about every isolation dimension. The same system may occupy several taxonomy categories. Figure 2 separates mechanism choice from the boundary that must be verified.

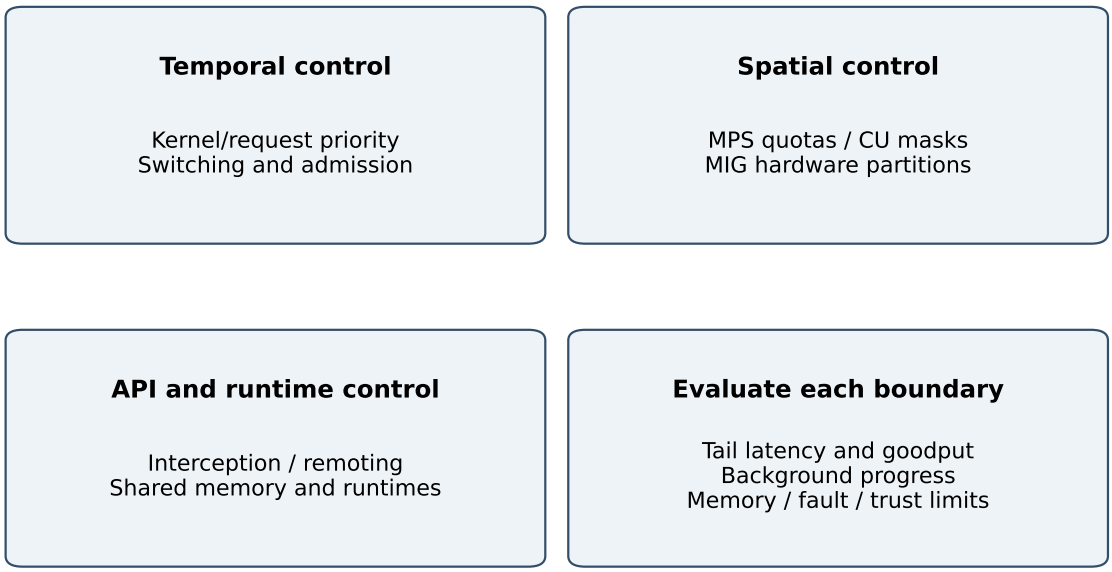


Figure 2: Mechanism and evaluation dimensions for the comparative corpus.

16 Comparative evidence

Numerical comparisons are within-paper contrasts. They are not ranked across unmatched GPUs, models, batches, load generators, memory capacities or baselines. Each row retains a representative outcome; the complete 52-row register is separate.

Report	Representative result	Comparator	Anchor /interpretation limit
P001	Throughput ratio: 2.7x	No MPS	pp. 457-458 Fig. 14 and accompanying prose; batch/rate/ quotas unresolved; not tail-latency evidence
P002	Prediction error range: 1.53-5.02 percent	Measured local latency	p. 822 section 5.2 Fig. 13; prediction accuracy, not utilization or latency speedup

Report	Representative result	Comparator	Anchor /interpretation limit
P003	Distribution threshold: 90% of measured pairs; 17.66 percent overhead below threshold	Solo-running latency	p. 203 Fig. 7; across-pair CDF, not request p90/p99; no hostile-memory/fault test
P004	Additional end-to-end latency: 6.01 ms	Ready-model latency subtracted; MPS overhead 307.02 ms	p. 509 Table 3; setup p. 508 section 6; mean of 100 runs; overhead, not total latency or p99; MPS uses unified-memory swapping
P005	Overall throughput ratio: 1.60x	RT-Only dedicated GPU	p. 549 section 7.2 Fig. 11 prose; setup p. 548 Table 2; includes RT and BE requests; RT-only reference dedicates GPU to RT tasks
P006	ResNet18 p99 reduction factor: up to 3x	CUDA-MS	printed p. 604 section 7.1/Fig. 12; PDF pp. 10-11; author-reported approximate maximum; not uniform across loads
P007	Aggregate throughput ratio: up to 7.3x	Dedicated high-priority inference	printed p. 1086 section 6.2.3/ Fig. 12; PDF p. 12; maximum; includes best-effort work
P008	GPU utilization efficacy (GUE): 94.04 composite score	Default MPS GUE 38.82	printed p. 500 Table 3; PDF p. 9; not percent GPU utilization
P009	Clockwork goodput: approximately 800 requests/second	INFaaS on nonidentical EC2 setup	printed p. 452 section 6.1/Fig. 5; author PDF p. 10; within-SLO successes; comparator qualitative; not total throughput
P010	Average P99 inference latency improvement factor, as reported: 12.6x	TaskSwitch (authors' adapted temporal-switch baseline)	printed p. 1666, section 6.1/ Figure 10 discussion; PDF p. 11; author-reported average across tested workloads; not a uniform per-model speedup, and not a comparison to unmodified PipeSwitch
P011	Reported throughput: 446 requests/s	same configuration with 1 process (354 requests/s)	author preprint PDF p. 5, section III-B prose; Figure 3 p. 4; one local configuration, not fleet savings. Source is arXiv v1; final published version not verified
P012	Reported single-GPU throughput factor: >10x	Native early binding at same offered function rate	p. 606 section 7.2; Figure 8 p. 605; PDF pp. 10-11; lower bound reported by authors, not an exact 10x; function count varies to saturate memory. No fleet cost claim

Report	Representative result	Comparator	Anchor /interpretation limit
P013	Author-reported average throughput factor: 6.7x	INFless with documented warm-up/ pipelined loading/ model sharing	p. 68 section 9.1/Figure 13; PDF p. 11; aggregate within-paper result across tested workflows; not universal or a matched cross-paper speedup. Batch/concurrency optimizations contribute
P014	Reported average QoS violation rate: 1.23 percent of requests	original solo P99 threshold (1% baseline violations by definition)	p. 1014 section 5.5/Figure 17; PDF p. 13; average across tested configurations; no claim that worst-case violation is 1.23%. Threshold-relative metric, not latency in milliseconds
P015	Reported high-priority P99 latency ratio: 1.07x standalone	standalone high-priority Bert-large execution	p. 683 section 7.4 case 3/Figure 15a; PDF p. 14; Triton+XSched; not a raw millisecond value. Native Triton ratio 1.53x; no NPU result included

Report	Additional comparative evidence	Interpretation boundary
P018 InferFair	524.29 versus 457.24 requests/s; unfairness 1.05 versus 1.50 at local load 3:1:2	SLO goodput and max/min model slowdown; not a p99 improvement.
P019 ArkGPU	All 30 tested inference/training pairs meet twice-solo p99 target; low-contention QPS can fall about 5 percent	Mixed workload evidence; preserve the regression and local/system distinction.
P020 Dyna-P	Local MIG concurrency and JCT sensitivity; predictor perturbation raises makespan	JCT and makespan do not establish request-tail SLO compliance.
P021 KRISP	Kernel-wise sizing evaluates throughput and p95 inference latency	AMD unrestricted concurrency baseline and emulation/overhead caveats apply.
P022 Tally	Background normalized throughput 1.19 versus TGS 1.65 at ten-percent idle-time setting	Foreground protection has a background-throughput cost.

Additional sources: [9, 10, 11, 12, 13].

17 Engineering trade-offs

Choice	Observed benefit /required sacrifice
Tuned MPS	VastPipe throughput contrasts do not establish tail-latency or fault isolation; iGniter recovery traces retain transient SLO violation.

Choice	Observed benefit /required sacrifice
Preemptive priority	XSched high-priority Triton P99 is 1.07x standalone versus 1.53x native Triton, with a continuously busy background client; background throughput remains a separate objective.
Memory handover	SIRIUS improves local inference tails versus an adapted switch baseline while opportunistic training uses residual resources; training gains are not inference throughput.
Workflow runtime sharing	StreamBox eliminates runtime redundancy but cross-user concurrency can cause queuing and SLO violations; within-workflow address-space sharing limits isolation.
Remoting/model swapping	Torpor increases low-rate function density; heavy model transfers can increase latency sharply and add CPU/bandwidth demand.
Temporal quota granularity	KRYPTON token lengths trade scheduling overhead against wait time, differently for different batches.
MIG+MPS configuration	ParvaGPU preprint example changes throughput from 354 to 446 requests/s while latency rises from 11 to 27 ms; this is version-qualified local evidence, not a final included-study claim.

ArkGPU's low-contention regression and Tally's lower background throughput show why foreground latency protection should be reported with the work sacrificed. A configuration that admits more concurrent work can still reduce useful per-tenant service. Dyna-P further separates partition-placement efficiency from online request latency: reducing job completion or fragmentation does not establish a p99 serving guarantee.

18 Conflicting evidence and comparability

Different results are not automatically contradictory. GSLICE reports a negative latency contrast against TensorRT in one configuration, while other outcomes evaluate different baselines and statistics. Torpor model swapping can hurt a heavy model even when function density improves. These within-paper differences support condition-aware choices rather than an averaged universal speedup.

The ParvaGPU local example illustrates a concrete tension: at 1-GPC MIG size and batch 4, moving from one to three MPS processes changes throughput from 354 to 446 requests/s and latency from 11 to 27 ms. The separately calculated increase is 25.99 percent in throughput and 2.4545x in latency, with explicit inputs retained. The latency percentile is unspecified and final publication equivalence is unresolved. These values are not pooled with P99/P98 request tails from other systems.

Cross-study comparisons remain qualitative where hardware, tenant mix, batching, load, runtime sharing or measurement intervals differ. No matched, controlled study-to-study contradiction or universal winning mechanism has been established by this checkpoint.

Table 16: Apparent conflicts and the reported conditions that explain them.

Studies compared	Apparent tension	Explaining condition
P001 vs P011	Throughput ratio 2.7x versus 25.99 percent (calculated) increase	Baseline (no MPS versus one MPS process), hardware and workload differ; metric is a ratio versus a calculated percentage.
P007 vs P015 vs P010	Aggregate throughput up to 7.3x versus high-priority P99 1.07x of standalone versus average P99 improvement 12.6x	Metric (aggregate throughput versus tail latency), baseline (dedicated, standalone, adapted switch) and workload differ.
P005 (REEF) vs P019	Overall throughput 1.60x versus about 5 percent lower QPS at low contention with p99 targets met	Workload (real-time plus best-effort versus inference/training pairs) and metric (throughput versus p99 target) differ.
P005, PyTorch vs TensorRT strata	Normalized p99/throughput relationship changes with inference engine	Baseline and implementation; strata are kept separate.
P008 (GSLICE)	Negative latency contrast against TensorRT in one configuration, other outcomes differ	Baseline and statistic differ within the paper.
P011 (ParvaGPU)	Throughput rises 354 to 446 requests/s while latency rises 11 to 27 ms	Same-configuration trade-off; percentile unspecified, so it is not pooled with P99/P98 tails.
P012 (Torpor)	Higher function density but heavy-model latency can worsen	Workload (model size) and memory transfer cost.
P018 (InferFair)	Goodput and fairness benefit varies	Workload (model-load ratio).

The journal REEF comparison is particularly informative: its reported normalized p99/throughput relationship changes with inference engine and implementation. PyTorch and TensorRT strata must remain separate rather than selecting only the most favorable point. InferFair likewise reports a local goodput/fairness benefit whose size varies with model-load ratio. These are condition-dependent results, not evidence that one mechanism dominates all workloads.

19 Answers to the research questions

RQ	Evidence-based checkpoint answer
RQ1	The assessed slice covers temporal scheduling, MPS, API interception/remoting, memory-sharing hybrids and MIG-integrating designs. Prevalence across the literature cannot be inferred from the incomplete selected set.

RQ	Evidence-based checkpoint answer
RQ2	The register distinguishes requests/s, throughput ratios, P99/P98, SLO-violation rate, API overhead and host/GPU memory. These are not interchangeable measures of utilization.
RQ3	Batch size, process count, arrival pattern, model size, runtime redundancy, hardware capability and memory/PCIe pressure alter outcomes. Local example settings accompany every metric.
RQ4	Scheduling fairness and resource quotas have more direct evidence than general security/fault guarantees. Compiler transformation, kernel modules, model swapping and CPU-side API handling impose different costs.
RQ5	Select a candidate mechanism by service priority and constraints, then validate it under the intended model mix and load. Final recommendation confidence is limited by incomplete eligibility, source versions and single-reviewer assessment.

20 Conditional engineering recommendations

Scenario	Candidate /evidence	Cost and alternative
Trusted multi-model inference with configurable quotas	Test MPS allocation approaches against native sharing on the target GPU; VastPipe/iGniter/ gpulets/GSLICE supply condition-specific leads.	Check tail latency and memory interference. Use stronger partitioning where trust boundaries require it; do not infer security from throughput.
Foreground request-tail priority	Test preemptive scheduling; the XSched Triton case and REEF/ Orion/Paella offer different integration choices.	Account for hardware preemption support and background slowdown; use batching/dedication when isolation/integration demands dominate.
Workflow-local function density	Test StreamBox-style runtime/ memory sharing where workflow functions share a trust boundary.	Requires compilation/profiling and controlled concurrency; separate runtimes/partitions suit stronger isolation requirements.
Many low-rate model functions	Test Torpor late binding against resident native baselines under target model sizes.	Host memory, CPU and PCIe traffic increase; keep heavy/ high-rate models resident when transfer cost dominates.
Mixed inference and opportunistic training	Test SIRIUS-type memory handover using separable inference SLO metrics.	Requires runtime integration and controlled memory reclamation; dedicate resources where interference risk is unacceptable.
MIG-aware deployments	Treat KRYPTON/ParvaGPU as design leads with different source/ validation status.	Verify supported profiles, reconfiguration limits and final source; hardware partitions alone do not establish a measured end-to-end security guarantee.

21 Research gaps

The comparative evidence lacks a common benchmark covering all mechanism families with identical GPU, software, model mix, load and baseline. Source reporting is uneven for repetitions, uncertainty intervals, runtime versions, energy and adversarial fault/security tests. The new interference and fairness observations motivate joint measurement of foreground service, background progress, memory/PCIe pressure and reconfiguration costs. Source-version and access differences constrain replication. The stopping-rule branch limits claims about prevalence and comprehensive coverage.

22 Threats to validity and limitations

Threat	Material limit and mitigation
Selection and coverage	This bounded rapid synthesis includes twenty studies. The stopping rule left 559 retained records outside source assessment, limiting claims about exhaustive recall or mechanism prevalence.
Review method	One consolidated decision pass was used for screening, appraisal and extraction. No inter-rater agreement statistic is reported; scores, evidence locations and provenance remain available.
Search reconstruction	Historical ACM export and ScienceDirect count/field discrepancies are preserved. Reconstructed exports do not establish undocumented historical clicks or facets.
Version and access	REEF companion identity is linked. ParvaGPU publication identity is verified, while detailed extraction uses its author manuscript. Source-version boundaries are retained in the evidence locations.
Measurements	No pooled speedup is calculated across unlike hardware, workloads or metrics. Within-paper baselines, scope and unfavorable results are retained.
Reporting availability	Printed extraction cards contain supported fields only; missing details remain missing in the audit and are not replaced by zero or guessed software versions.

The engineering recommendations apply to the observed conditions and require validation on the intended deployment. The original submitted protocol is reproduced unchanged as submitted; departures from it, including the single-reviewer implementation. All revisions are reversible through stable report/study/outcome IDs and the accompanying data files.

23 Conclusion

The twenty-study comparative corpus supports a conditional engineering choice. Temporal preemption suits foreground request priority when switching and integration costs are tolerable. MPS-style sharing improves density in trusted workloads but requires interference-aware quotas and workload-specific validation. MIG adds hardware partition boundaries while imposing placement and reconfiguration constraints. API interception and shared runtimes reduce selected deployment or memory costs with compatibility and trust trade-offs. No source supports a universal winner across hardware, model mixes and isolation

requirements. The evidence register preserves the cost as well as the benefit of each choice. The discovery queue remains outside the bounded source-assessed synthesis.

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A Paper by paper quality and extraction evidence

Each card joins one study ID to its report ID. N/A denotes an explicitly justified inapplicable quality criterion. Printed extraction fields are limited to supported information; source-specific limits accompany each card. These tables do not create additional outcomes or studies.

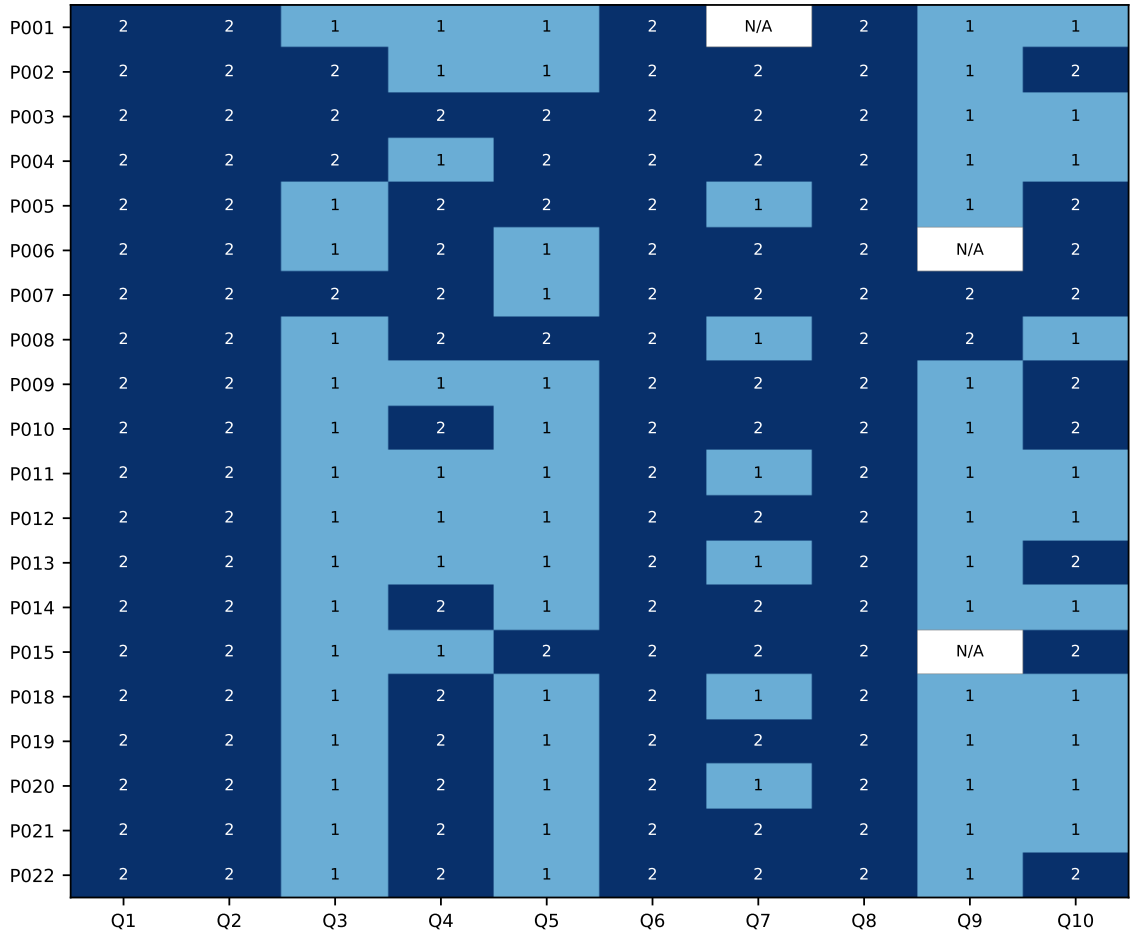


Figure 3: Q1–Q10 appraisal matrix for the twenty-study corpus. Report IDs P016 and P017 are absent because those reports were assessed at full text and excluded (EC2), so they were not quality-scored.

A.1 S001 VastPipe

Report P001. Source: `https://doi.org/10.1007/s11390-024-3773-5`.

Quality assessment. 13/18 applicable points (72.2 percent).

Item	Score	Rationale	Evidence location
Q1	2	Objective and local GPU portability question explicit	Published pp. 457-458; PDF pp. 16-17; Sections 6.7-6.9; Figures 14-16
Q2	2	Tuned MPS quotas and CU masking described	Published pp. 458-459; PDF pp. 16-17; Sections 6.7-6.9; Figures 14-16
Q3	1	GPU-specific repeats and variance unresolved in conversion	Published pp. 458-459; PDF pp. 16-17; Sections 6.7-6.9; Figures 14-16
Q4	1	GPU/backend named; precise GPU loads and versions incomplete	Published pp. 458-459; PDF pp. 16-17; Sections 6.7-6.9; Figures 14-16
Q5	1	Same hardware and no/default/tuned MPS; quotas incomplete	Published pp. 458-459; PDF pp. 16-17; Sections 6.7-6.9; Figures 14-16
Q6	2	Throughput ratios and baseline named in prose	Published pp. 458-459; PDF pp. 16-17; Sections 6.7-6.9; Figures 14-16
Q7	NA	GPU-portability extraction makes no latency claim; do not score NPU latency as GPU evidence	Published pp. 458-459; PDF pp. 16-17; Sections 6.7-6.9; Figures 14-16
Q8	2	Multiple named co-located inference pairs evaluated	Published pp. 458-459; PDF pp. 16-17; Sections 6.7-6.9; Figures 14-16
Q9	1	Performance interference considered; no separate security/fault validation extracted	Published pp. 458-459; PDF pp. 16-17; Sections 6.7-6.9; Figures 14-16
Q10	1	Stable-mix Q-table constraint; GPU replication settings incomplete	Published pp. 458-459; PDF pp. 16-17; Sections 6.7-6.9; Figures 14-16

Structured extraction.

Field	Source-supported content
Authors	Li-Xian Ma and Le-Ping Wang and En Shao and Rong-Yu Cao and Guang-Ming Tan
Year	2025
Venue	Journal of Computer Science and Technology
DOI	10.1007/s11390-024-3773-5
Mechanism	MPS adaptive quotas; separate AMD CU-mask stratum
Hardware and environment	NVIDIA A100 for MPS; AMD MI100 for CU-mask tests
Runtime details	NVIDIA TensorRT serving; AMD TVM/ROCm with hipExtStreamCreateWithCUMask

Field	Source-supported content
Workload	CV pairs and BERT/GPT2 mixed model pairs; Poisson-generated request mixes
Tenants	Two co-located inference model services in reported local model-pair comparisons
Comparator	No MPS; default MPS without per-service quotas; RL-selected MPS quotas. AMD no/default CU mask separately.
Overhead	Offline Q-table generation; plans reused within request windows; source Section 6.9
Limitations	Include only local GPU inference evidence, not the main 1000-NPU cluster results. Offline Q-table assumes relatively stable request mix; GPU-specific quotas, repetitions and variance unresolved. Chart bars are not preserved in Markdown. Related ICCD 2021 report is linked as a companion-version lead outside the synthesis corpus.
Source-linked finding	A100 VGG16+GoogLeNet: throughput 2.7x no MPS and 1.5x default MPS; not a universal average or p99 result. NLP/CV mean 2.43x no MPS, 1.45x default MPS. MI100 mean 1.89x no mask, 1.24x default mask.
Performance isolation	Throughput/interference comparison only; no extracted tail-latency or SLO protection measurement.
Security boundary	Not established
Fault boundary	Not tested in extracted GPU results

Evidence location: Published pp. 457-458; PDF pp. 16-17; Sections 6.7-6.9; Figures 14-16.

Registered evidence rows.

Evidence ID	Metric and observation	Conditions and interpretation
E001	Throughput ratio: 2.7 x	VGG16+GoogLeNet; baseline: No MPS. Batch/rate/quotas unresolved; not tail-latency evidence Source: pp457-458 Fig14 and accompanying prose

Evidence ID	Metric and observation	Conditions and interpretation
E002	Throughput ratio: 1.5 x	VGG16+GoogLeNet; baseline: Default MPS. Same comparison family; not a universal average Source: pp457-458 Fig14 and accompanying prose

A.2 S002 iGniter

Report P002. Source: `https://doi.org/10.1109/TPDS.2022.3232715`.

Quality assessment. 17/20 applicable points (85.0 percent).

Item	Score	Rationale	Evidence location
Q1	2	Interference-aware inference provisioning objective explicit	abstract,section1
Q2	2	MPS allocation and shadow implementation described	section4.2
Q3	2	Model checks repeated three times with SD; metric windows specified	sections5.1-5.3
Q4	1	GPU/workloads/backend detailed; OS/driver unresolved	sections4.2,5.1
Q5	1	Adapted baselines and allocation/batch differences disclosed	section5.1
Q6	2	Cost, throughput, resource-allocation metrics distinguished	sections5.1,5.3
Q7	2	P99 windows, SLO and recovery timing explicit	section5.3 Fig17
Q8	2	Four co-located models and quota configuration specified	section5.2 Fig13
Q9	1	Performance interference measured; security/fault boundaries not tested in checked portion	sections5.2-5.3
Q10	2	Code, prediction limitations and PCIe caveat explicit	sections4.2,5.2

Structured extraction.

Field	Source-supported content
Authors	Fei Xu and Jianian Xu and Jiabin Chen and Li Chen and Ruitao Shang and Zhi Zhou and Fangming Liu
Year	2023
Venue	IEEE Transactions on Parallel and Distributed Systems
DOI	10.1109/TPDS.2022.3232715
Mechanism	MPS active-thread quotas + interference-aware batch/provisioning and shadow process
Hardware and environment	Ten EC2 p3.2xlarge instances,one V100 each,8vCPU/61GB; Triton2.12.0,TensorRT8.0.1.6. MPS active-thread percentage; allocation unit2.5%. Heterogeneous extension adds20 T4 instances.
Workload	AlexNet,ResNet50,VGG19(ImageNet),SSD(VOC2012);12 workload/SLO configurations; constant arrivals. Local experiment co-locates four models at25% quota,batch3. p99 provisioning measured over windows e.g.30s; model-validation experiments repeated3 times with SD bars.
Batch load and SLO	batch3 for prediction check;W1 SLO10ms for recovery example
Comparator	FFD+; GSLICE+ with iGniter placement; gpu-lets+ modified batch strategy and best-fit placement. These are adapted baselines.

Field	Source-supported content
Limitations	Constant arrivals and four vision models limit bursty/LLM transfer. Negligible PCIe contention assumption can underpredict AlexNet; baselines modified. Steady-state/post-correction SLO success is not an unconditional guarantee; shadow standby incurs capacity/memory costs. Chart bars/latency traces unavailable in Markdown.
Source-linked finding	Local four-model latency prediction error1.53%-5.02%(Fig13). Fig14 deployment: iGniter6 GPUs/0 SLO violations vs gpu-lets+8/3,FFD+5/10,GSLICE+6/3;25% lower hourly instance cost vs gpu-lets+, not25% higher local utilization. Fig17 shows an initial p99 violation before shadow-process recovery.
Performance isolation	Local interference prediction and p99/SLO checks; transient prediction failure explicitly retained
Security boundary	Not established
Fault boundary	Not tested in extracted results

Evidence location: Published pp820-824; sections4.2,5.1-5.4; Figs13-18; Markdown lines938-1245.

Registered evidence rows.

Evidence ID	Metric and observation	Conditions and interpretation
E003	Prediction error range: 1.53-5.02 percent	AlexNet/ResNet50/VGG19/SSD; batch3; baseline: Measured local latency. Prediction accuracy, not utilization or latency speedup Source: p822 section5.2 Fig13

Evidence ID	Metric and observation	Conditions and interpretation
E004	Initial p99 latency: 15.6 ms	W1 with W5 and W11; baseline: W1 latency SLO10ms. Transient violation precedes recovery; no unconditional guarantee Source: p823 section5.3 Fig17 prose

A.3 S003 gpulets

Report P003. Source: `https://www.usenix.org/system/files/atc22-choi-seungbeom.pdf`.

Quality assessment. 18/20 applicable points (90.0 percent).

Item	Score	Rationale	Evidence location
Q1	2	Clear SLO-aware co-location objective	p199 abstract;section1
Q2	2	MPS partitions and temporal batching described	pp201-207 sections2-4
Q3	2	Poisson requests;median of3 repeats;throughput/SLO procedure	pp207-208 section5.1-5.2
Q4	2	Full GPU/CPU/OS/driver/runtime and model tables	p207 Tables2-3
Q5	2	Ported SBP comparator and Nexus incompatibility justified	p207 section5.1
Q6	2	Maximum throughput with99%SLO definition;system scale explicit	p208 section5.2
Q7	2	SLO99% procedure;local overhead CDF population explicit	p203 Fig7;p208 section5.2
Q8	2	Paired model interference and scheduler interference comparison	p203 Fig7;p208 Fig14
Q9	1	Performance interference checked;security/fault boundary not established in the checked source evidence	p203 Fig7;section5.2
Q10	1	Prototype and appendix setup available;replication not attempted	p207 Table2;appendixA

Structured extraction.

Field	Source-supported content
Authors	Seungbeom Choi et al.
Year	2022
Venue	USENIX ATC22
DOI	not resolved
Mechanism	MPS + temporal
Hardware and environment	RTX2080Ti 11GB; two/server; Ubuntu18.04; CUDA10.2; driver440.64; PyTorch1.10
Runtime details	NVIDIA440.64;CUDA10.2;PyTorch1.10 (p207 Table2)
Workload	Nine models; Poisson arrivals; per-model SLO
Tenants	paired local inferences; system scenarios contain several models
Batch load and SLO	Poisson inter-arrivals; model SLO5-202ms;99% requests within SLO; median of3 runs (pp207-208)
Comparator	Adapted SBP/Nexus; best-fit with/without interference
Latency scope	Fig7 across-pair CDF threshold17.66%; not request-p90
Scalability	8 logical backend servers with additional GPUs; system-level result (pp208-209)
Overhead	20-second reorganization interval; overlap new proxy preparation with active serving (p207 Fig11)
Limitations	Server-level gains cannot be treated as one-GPU utilization.

Field	Source-supported content
Trade-offs	Partition and batch jointly selected; throughput numbers at system scale cannot be assigned to one GPU
Source-linked finding	Local gpulets support interference-aware provisioning; cluster throughput is a different outcome.

Evidence location: PDF p10 Table2/3; section5; local profiling section3.

Registered evidence rows.

Evidence ID	Metric and observation	Conditions and interpretation
E006	Distribution threshold:90% of measured pairs: 17.66 percent overhead below threshold	Paired model inferences; baseline: Solo-running latency. Across-pair CDF, not request-p90/p99; no hostile-memory/fault test Source: p203 Fig7

A.4 S004 PipeSwitch

Report P004. Source: `https://www.usenix.org/system/files/osdi20-bai.pdf`.

Quality assessment. 17/20 applicable points (85.0 percent).

Item	Score	Rationale	Evidence location
Q1	2	Fine-grained task-switching problem and design explicit	p499 abstract;section1
Q2	2	Controller,memory daemon,workers and pipeline described	pp504-508 sections4-5
Q3	2	100-run mean;min/max forFig6;setup described	p508 section6
Q4	1	Hardware/software specified;OS/kernel/driver not established in the checked source evidence	p508 section6
Q5	2	Same modified PyTorch;ready-model and swapping conditions explicit	p508 section6.1
Q6	2	Inference throughput and utilization denominator defined	p509 section6.1 Fig6
Q7	2	End-to-end overhead defined;mean and extrema identified	pp508-509 Table3
Q8	2	Training/inference co-location;inference metric separated	pp508-509 section6.1
Q9	1	Process/memory control described;security/fault testing not established in the checked source evidence	pp507-508 implementation
Q10	1	Versioned stack described;limitations documented;replication not attempted	p508 section6;section7

Structured extraction.

Field	Source-supported content
Authors	Zhihao Bai;Zhen Zhang;Yibo Zhu;Xin Jin
Year	2020
Venue	USENIX OSDI20
DOI	not resolved
Mechanism	Temporal context switching
Hardware and environment	V10016GB or T416GB EC2; PyTorch1.3; CUDA10.1
Runtime details	CUDA10.1;PyTorch1.3.0;torchvision0.4.2;scipy1.3.2;
Workload	Training batch32; inference8; RN152/InceptionV3/BERT; 100 runs
Tenants	one training task preempted for one inference task; separable inference latency
Batch load and SLO	training32;inference8;100-run averages;Fig6 reports minimum/maximum (p508)
Comparator	Unoptimized/grouped/per-layer transfer
Latency scope	Additional total latency versus ready model;Table3;not p99
Overhead	Table3 total extra latency;Table4 first-layer startup overhead;kept separate
Advantages	Reduced switching overhead for checked workloads; see extraction.csv

Field	Source-supported content
Limitations	Average switch/inference latency is not p99; multi-VM throughput separate.
Trade-offs	Pipelined transfers and registered resident host models require runtime integration; eligibility is based on separable inference-preemption latency in Table 3.
Source-linked finding	Fig7: inference preemption latency reduced by up to 38.2ms.

Evidence location: PDF pp11-13; sections6-6.2; Fig7.

Registered evidence rows.

Evidence ID	Metric and observation	Conditions and interpretation
E005	Additional end-to-end latency: 6.01 ms	ResNet152 inference batch8 + training batch32; baseline: Ready-model latency subtracted; MPS overhead307.02ms. Mean100 runs; overhead, not total latency or p99; MPS uses unified-memory swapping Source: p509 Table3; setup p508 section6

Evidence ID	Metric and observation	Conditions and interpretation
E007	Additional end-to-end latency: 307.02 ms	ResNet152 inference batch8 + training batch32; baseline: Ready-model latency subtracted. Training task occupies GPU memory and continues. Baseline condition matters; not a universal MPS comparison. Source: p509 Table3; setup p508 section6

Evidence ID	Metric and observation	Conditions and interpretation
E008	Additional end-to-end latency: 5.40 ms	Inception_v3 inference batch8 + training batch32; baseline: Ready-model latency subtracted. Mean of100 runs; overhead not total latency or tail statistic; mixed training/inference workload. Source: p509 Table3; setup p508 section6

Evidence ID	Metric and observation	Conditions and interpretation
E009	Additional end-to-end latency: 232.25 ms	Inception_v3 inference batch8 + training batch32; baseline: Ready-model latency subtracted. Training task occupies GPU memory and continues. Baseline condition matters; not a universal MPS comparison. Source: p509 Table3; setup p508 section6

Evidence ID	Metric and observation	Conditions and interpretation
E010	Additional end-to-end latency: 10.27 ms	Bert_base inference batch8 + training batch32; baseline: Ready-model latency subtracted. Mean of100 runs; overhead not total latency or tail statistic; mixed training/inference workload. Source: p509 Table3; setup p508 section6

Evidence ID	Metric and observation	Conditions and interpretation
E011	Additional end-to-end latency: 204.52 ms	Bert_base inference batch8 + training batch32; baseline: Ready-model latency subtracted. Training task occupies GPU memory and continues. Baseline condition matters; not a universal MPS comparison. Source: p509 Table3; setup p508 section6

Evidence ID	Metric and observation	Conditions and interpretation
E012	Additional end-to-end latency: 5.57 ms	ResNet152 inference batch8 + training batch32; baseline: Ready-model latency subtracted. Mean of100 runs; overhead not total latency or tail statistic; mixed training/inference workload. Source: p509 Table3; setup p508 section6

Evidence ID	Metric and observation	Conditions and interpretation
E013	Additional end-to-end latency: 259.20 ms	ResNet152 inference batch8 + training batch32; baseline: Ready-model latency subtracted. Training task occupies GPU memory and continues. Baseline condition matters; not a universal MPS comparison. Source: p509 Table3; setup p508 section6

Evidence ID	Metric and observation	Conditions and interpretation
E014	Additional end-to-end latency: 7.66 ms	Inception_v3 inference batch8 + training batch32; baseline: Ready-model latency subtracted. Mean of100 runs; overhead not total latency or tail statistic; mixed training/inference workload. Source: p509 Table3; setup p508 section6

Evidence ID	Metric and observation	Conditions and interpretation
E015	Additional end-to-end latency: 193.05 ms	Inception_v3 inference batch8 + training batch32; baseline: Ready-model latency subtracted. Training task occupies GPU memory and continues. Baseline condition matters; not a universal MPS comparison. Source: p509 Table3; setup p508 section6

Evidence ID	Metric and observation	Conditions and interpretation
E016	Additional end-to-end latency: 34.56 ms	Bert_base inference batch8 + training batch32; baseline: Ready-model latency subtracted. Mean of100 runs; overhead not total latency or tail statistic; mixed training/inference workload. Source: p509 Table3; setup p508 section6

Evidence ID	Metric and observation	Conditions and interpretation
E017	Additional end-to-end latency: 338.25 ms	Bert_base inference batch8 + training batch32; baseline: Ready-model latency subtracted. Training task occupies GPU memory and continues. Baseline condition matters; not a universal MPS comparison. Source: p509 Table3; setup p508 section6

A.5 S005 REEF conference

Report P005. Source: `https://www.usenix.org/system/files/osdi22-han.pdf`.

P023 identifies the journal companion. The conference report is retained as the source for the original quality items and conference outcomes [28, 29]. Journal extension observations use the version noted on this card; both reports map to one study.

Quality assessment. 17/20 applicable points (85.0 percent).

Item	Score	Rationale	Evidence location
Q1	2	Concurrent real-time/best-effort inference objective explicit	p539 abstract;section1
Q2	2	Reset preemption,queues and padding implementation described	sections3-6;pp547-548
Q3	1	Benchmark arrivals and baseline methods detailed;repeat/variance protocol not established in the checked source evidence	p548 section7.1
Q4	2	CPU,GPU,host RAM,OS,runtime,models andclients specified	p548 section7.1 Table2
Q5	2	RT-only and GPUStreams/SEQ conditions;extensions disclosed	p548 section7.1;p549 section7.3
Q6	2	Total RT+BE throughput and denominator explicit	p549 section7.2 Fig11
Q7	1	End-to-end latency and load specified;request percentile not established in the checked source evidence	pp548-549 Table2/Fig11
Q8	2	MultipleRT/BE clients and multiple mixes directly tested	p548 Table2;p549 section7.2
Q9	1	Performance isolation tested;no extracted memory/security/fault boundary test	pp548-549 section7.2
Q10	2	Artifact and idempotence/platform limitations explicit	p540 repository;section8;appendixA

Structured extraction.

Field	Source-supported content
Authors	Mingcong Han;Hanze Zhang;Rong Chen;Haibo Chen
Year	2022
Venue	USENIX OSDI22
DOI	not resolved
Mechanism	Reset-based temporal preemption + dynamic kernel padding; NVIDIA binary instrumentation
Hardware and environment	Conference: AMD MI50, 16GB, 60 CUs. Journal companion: MI50 16GB and separate NVIDIA GV100 32GB/80 SMs. Version-specific platforms are not a common configuration.
Runtime details	ROCm4.3.0;ApacheTVM0.8.0;NVIDIA extension CUDA10.2
Workload	DISB A-E,uniform RT except Poisson E; closed-loop BE; ResNet152,VGG19,YOLOv3,BERT-base,GPT2 (128 tokens,GPT2 prefill). RT rates50%,100% or10% solo-throughput according to workload; Apollo/SVL trace. Separate LLM case batch1,100 decode tokens,MLC-LLM.

Field	Source-supported content
Tenants	DISB A/B:1RT+1BE;C:1RT+5BE;D/E:5RT+5BE
Batch load and SLO	Table2 RT100/220/100/20/20req/s;D uniform,E Poisson;BE closed-loop;
Comparator	RT-only; sequential priority; GPU streams. NVIDIA closed-source strata compare priority-enabled Triton/TensorRT and Orion/PyTorch without dynamic padding.
Latency scope	RT end-to-end overhead below2% in REAL;no request percentile extracted
Throughput scope	RT+BE total normalized to RT-Only;not RT-only throughput
Advantages	Checked workloads retain near RT-only latency while using idle capacity for BE inference
Limitations	Conference outcomes and original QA remain OSDI 2022 anchored; journal observations are explicitly labelled companion results. Deterministic/stateless kernel assumption; closed-source comparison omits dynamic padding. Sufficient memory and bandwidth assumptions constrain transfer. Both versions map to one study identity.
Trade-offs	DISB D/E throughput about25% below GPUStreams while limiting RT latency;implementation differs on closed-source GPUs
Source-linked finding	GV100 PyTorch DISB-C: Orion p99 normalized latency1.17x vs Reef below1.05x; normalized total throughput0.97x vs1.14x(Fig29). TensorRT comparison trades throughput: mean1.21x Triton vs1.11x Reef(Fig28). All ratios retain model-solo normalization; do not mix engines/AMD strata.
Performance isolation	Per-request normalized RT distributions/p99 and separately normalized total throughput; engine/stratum retained
Security boundary	Not established by scheduler priority
Fault boundary	Kernel killing/restart is a scheduling mechanism, not adversarial fault containment
Report version	Journal 10.1145/3768622 is an extension of OSDI 2022 REEF; both report versions map to S005 once. Earlier conference outcomes remain conference-anchored.
Companion report ID	P023
Companion DOI	10.1145/3768622

Evidence location: Conference: printed pp. 548-549, Sections 7.1-7.2; journal extension: article pp. 1:23-1:24 and 1:31-1:34, Figures 28-29.

Registered evidence rows.

Evidence ID	Metric and observation	Conditions and interpretation
E018	Overall throughput ratio: 1.60 x	DISB A:1 VGG19 RT client100req/s uniform +1 ResNet152 BE; baseline: RT-Only dedicated GPU. Includes RT and BE requests; RT-only reference dedicates GPU to RT tasks. Source: p549 section7.2 Fig11 prose; setup p548 Table2

Evidence ID	Metric and observation	Conditions and interpretation
E019	RT end-to-end latency overhead: 0.5 percent	DISB A:1 VGG19 RT client100req/s uniform +1 ResNet152 BE; baseline: RT-Only dedicated GPU. Described as negligible; not a p99 value or worst-case guarantee. Source: p549 section7.2 Fig11 prose; setup p548 Table2

Evidence ID	Metric and observation	Conditions and interpretation
E020	Overall throughput ratio: 1.14 x	DISB B:1 VGG19 RT client220req/s uniform +1 ResNet152 BE; baseline: RT-Only dedicated GPU. High RT load; differs from low-load DISB A. Source: p549 section7.2 Fig11 prose; setup p548 Table2

Evidence ID	Metric and observation	Conditions and interpretation
E021	RT end-to-end latency overhead: 1 percent	DISB B:1 VGG19 RT client220req/s uniform +1 ResNet152 BE; baseline: RT-Only dedicated GPU. About60 microseconds in prose; not a tail percentile. Source: p549 section7.2 Fig11 prose; setup p548 Table2

Evidence ID	Metric and observation	Conditions and interpretation
E022	RT end-to-end latency overhead: <1.5 percent	DISB C-E:multiple BE clients; baseline: RT-Only dedicated GPU. Authors also state0.1ms; keep inequality and scenario range. Source: p549 section7.2 Fig11 prose; setup p548 Table2

Evidence ID	Metric and observation	Conditions and interpretation
E023	Overall throughput ratio: 3.00 x	DISB D:5 RT model clients20req/s uniform +5 BE model clients; baseline: RT-Only dedicated GPU. About25% lower throughput than GPUStreams; latency trade-off preserved. Source: p549 section7.2 Fig11 prose; setup p548 Table2

Evidence ID	Metric and observation	Conditions and interpretation
E024	Overall throughput ratio: 2.96 x	DISB E:5 RT model clients20req/s Poisson +5 BE model clients; baseline: RT-Only dedicated GPU. About25% lower throughput than GPUStreams; latency trade-off preserved. Source: p549 section7.2 Fig11 prose; setup p548 Table2

Evidence ID	Metric and observation	Conditions and interpretation
E025	Overall throughput ratio: 7.7 x	Apollo REAL trace about43 RTreq/s +5 BE model clients; baseline: RT-Only dedicated GPU. Real-world trace mapped to representative DNN models; not production deployment measurement. Source: p549 section7.2 Fig11 prose; setup p548 Table2

Evidence ID	Metric and observation	Conditions and interpretation
E026	RT end-to-end latency overhead: <2 percent	Apollo REAL trace about43 RTreq/s +5 BE model clients; baseline: RT-Only dedicated GPU. Not a p99 latency claim or security/fault isolation test. Source: p549 section7.2 Fig11 prose; setup p548 Table2

A.6 S006 Paella: Low-latency Model Serving with Software-defined GPU Scheduling

Report P006. Source: `https://vincen.tl/files/ng23paella.pdf`.

Quality assessment. 16/18 applicable points (88.9 percent).

Item	Score	Rationale	Evidence location
Q1	2	Explicit goal	p595 section1
Q2	2	Mechanism classifiable	pp599-603 sections4-6
Q3	1	Repetition count unclear	p604 methodology
Q4	2	Versioned setup	p604 methodology
Q5	1	Wrapper caveat	p604 Table3/baselines
Q6	2	Defined throughput	pp604-605 Figs11-12
Q7	2	p99 specified	pp604-605 Figs11-12
Q8	2	Mixed jobs tested	p605 Figs12-13
Q9	NA	No security/fault or resource-isolation guarantee appraised; fairness is assessed under Q8; N/A with stated rationale	pp606-607 discussion;scope-specific rationale
Q10	2	Artifact and limits	pp606-607 section8

Structured extraction.

Field	Source-supported content
Authors	Kelvin K.W. Ng; Henri Maxime Demoulin; Vincent Liu
Year	2023
Venue	SOSP 2023
DOI	10.1145/3600006.3613163
Mechanism	Temporal software scheduling + MPS comparison
Hardware and environment	TeslaT4; XeonSilver4114/64GB; Linux4.15.0; CUDA11.7; modified TVM0.10
Runtime details	Linux4.15.0;CUDA11.7;TVM0.10.0
Workload	Eight models; lognormal sigma1.5/2; stabilized measurements
Tenants	Mixed inference; lognormal arrivals sigma1.5/2
Comparator	CUDA single/multiple streams; MPS; Triton23.03; ablations
Limitations	Custom Triton TVM wrapper; MPS omitted from Fig11; dynamic batching unsupported.
Trade-offs	Fairness versus short-job latency; instrumentation cost
Source-linked finding	p99 latency and throughput under bursty model mixes; Fig12 includes MPS.

Evidence location: PDF pp9-12; section7 Table3/Figs11-12; section8.

Registered evidence rows.

Evidence ID	Metric and observation	Conditions and interpretation
E027	ResNet18 p99 reduction factor: up to3 x	ResNet18+InceptionV3;lognormal sigma1.5/2; baseline: CUDA-MS. Author-reported approximate maximum; not uniform across loads Source: printed p604 section7.1/Fig12; PDF pp10-11

Evidence ID	Metric and observation	Conditions and interpretation
E028	p90 instrumentation overhead: 6.6 microseconds	Empty kernel;160 blocks;instrumentation with aggregation; baseline: Uninstrumented kernel. Microbenchmark; not inference request latency Source: printed p606 section7.3/Fig15; PDF p12

A.7 S007 Orion: Interference-aware, Fine-grained GPU Sharing for ML Applications

Report P007. Source: `https://fotstrt.github.io/files/2024-orion.pdf`.

Quality assessment. 19/20 applicable points (95.0 percent).

Item	Score	Rationale	Evidence location
Q1	2	Explicit goal	p1075 section1
Q2	2	Interception specified	pp1079-1082 sections4-5
Q3	2	Three repetitions	p1082 section6.1
Q4	2	Versioned setup	p1082 section6.1
Q5	1	Reimplemented comparator	p1083 baselines
Q6	2	Defined aggregate metric	p1086 section6.2.3
Q7	2	p99 and arrivals	p1086 section6.2.3
Q8	2	Collocations tested	p1086 section6.3
Q9	2	Trust boundary explicit	p1087 section7 Security
Q10	2	Artifact and limitations	p1087 section7;artifact appendix

Structured extraction.

Field	Source-supported content
Authors	Foteini Strati; Xianzhe Ma; Ana Klimovic
Year	2024
Venue	EuroSys 2024
DOI	10.1145/3627703.3629578
Mechanism	CUDA API interception + temporal
Hardware and environment	V10016GB/CUDA10.2; A10040GB/CUDA11.3; PyTorch1.12; Python3.9
Runtime details	PyTorch1.12;Python3.9;CUDA10.2/11.3
Workload	Inference-inference and inference-training; uniform/Poisson; three repeats
Tenants	High-priority plus best-effort inference;Poisson
Comparator	MPS; streams; temporal; REEF-N reimplementation
Limitations	Prototype clients share a process; REEF-N differs from AMD REEF.
Trade-offs	Tail latency versus best-effort progress
Source-linked finding	Interference-aware scheduling compares inference latency and BE throughput.

Evidence location: PDF pp8-9; sections5.3-6.1; Tables1/3.

Registered evidence rows.

Evidence ID	Metric and observation	Conditions and interpretation
E029	Aggregate throughput ratio: up to7.3 x	Two inference clients;Poisson;V100; baseline: Dedicated high-priority inference. Maximum; includes best-effort work Source: printed p1086 section6.2.3/Fig12; PDF p12

Evidence ID	Metric and observation	Conditions and interpretation
E030	High-priority p99 overhead: 15 percent	Two inference clients;Poisson;V100; baseline: Dedicated high-priority inference. Authors summarize within15%; averaged across collocations, not worst-case guarantee Source: printed p1086 section6.2.3/Fig12; PDF p12

A.8 S008 GSLICE: Controlled Spatial Sharing of GPUs for a Scalable Inference Platform

Report P008. Source: `https://www.cse.iitb.ac.in/~puru/courses/spring2024-25/downloads/gslice.pdf`.

Quality assessment. 17/20 applicable points (85.0 percent).

Item	Score	Rationale	Evidence location
Q1	2	Explicit goal	p492 introduction
Q2	2	Sharing described	pp495-499 section3
Q3	1	Confidence intervals; counts unclear	p500 error bars;section4
Q4	2	Hardware and OS	pp499-500 testbed
Q5	2	Comparator scope stated	p501 section4.3
Q6	2	GUE defined	p494 GUE definition;p500 Table3
Q7	1	Tail statistic unclear	pp500-501 latency plots
Q8	2	Concurrent functions	p500 section4.2
Q9	2	Fixed quotas and performance-isolation tests; no stronger security claim inferred	pp496-498 resource controls;p502 performance isolation
Q10	1	Reproduction incomplete	pp499-504 implementation/evaluation/limits

Structured extraction.

Field	Source-supported content
Authors	Aditya Dhakal; Sameer G Kulkarni; K.K. Ramakrishnan
Year	2020
Venue	SoCC 2020
DOI	10.1145/3419111.3421284
Mechanism	MPS adaptive provisioning
Hardware and environment	One V10016GB; XeonGold6148; kernel4.4.0-150; DPDK19.02; CNTK/PyTorch/TensorRT
Runtime details	Linux4.4.0-150;DPDK19.02;CNTK/PyTorch/TensorRT
Workload	Two-four inference functions; VGG case SLO50ms; traffic replay
Tenants	Concurrent inference functions;25ms target
Comparator	Standalone functions; MPS provisioning variants
Limitations	GUE is not GPU busy percentage; median latency not p99; academic mirror.
Trade-offs	Batching/fairness versus individual latency
Source-linked finding	GUE measures efficiency; shadow instance loading takes about12s in Fig15.

Evidence location: PDF pp8-12; section4; Table3; Fig15.

Registered evidence rows.

Evidence ID	Metric and observation	Conditions and interpretation
E031	GPU utilization efficacy GUE: 94.04 composite score	Alexnet+ResNet50;25ms target; baseline: Default MPS GUE38.82. Not percent GPU utilization Source: printed p500 Table3; PDF p9

Evidence ID	Metric and observation	Conditions and interpretation
E032	Signed latency improvement: -7.52 percent	ResNet50;batch1;GPU-ready batch execution; baseline: TensorRT server19.02. Negative means worse latency; excludes transport overhead; single-model comparison Source: printed p501 Table4; PDF p10

A.9 S009 Serving DNNs like Clockwork: Performance Predictability from the Bottom Up

Report P009. Source: `https://www.usenix.org/system/files/osdi20-gujarati.pdf`.

Quality assessment. 16/20 applicable points (80.0 percent).

Item	Score	Rationale	Evidence location
Q1	2	Explicit goal	p443 introduction
Q2	2	Scheduling described	pp448-451 sections4-5
Q3	1	Repetition count unclear	p451 section6 setup
Q4	1	Versions not established in the checked source evidence	p451 section6 setup
Q5	1	Hardware mismatch disclosed	p451 section6.1 mismatch
Q6	2	Goodput defined	p452 goodput definition
Q7	2	SLO and CDF	pp451-452 Fig5
Q8	2	Contending models	p451 section6.1
Q9	1	No fault guarantee	pp455-456 discussion
Q10	2	Artifact and limits	pp455-456 discussion;artifact appendix

Structured extraction.

Field	Source-supported content
Authors	Arpan Gujarati; Reza Karimi; Safya Alzayat; Wei Hao; Antoine Kaufmann; Ymir Vigfusson; Jonathan Mace
Year	2020
Venue	USENIX OSDI 2020
Mechanism	Temporal exclusive scheduling
Hardware and environment	V10032GB; DellR740; worker32cores/768GB; two GPUs/host
Workload	Local experiment: one GPU; 15 RN50 copies; 16 clients/model; SLO10-500ms
Tenants	15 ResNet50 instances;16 closed-loop clients/model
Comparator	Clockwork variants; INFaaS on different hardware
Limitations	Cross-hardware INFaaS comparison is indirect; controller/worker distinction.
Trade-offs	Predictability versus admission and concurrency
Source-linked finding	Fig5 measures goodput satisfying SLO, rather than raw request throughput.

Evidence location: PDF p10; section6.1; Fig5.

Registered evidence rows.

Evidence ID	Metric and observation	Conditions and interpretation
E033	Clockwork goodput: approximately800 requests/second	15 ResNet50 instances;16 clients/model;500ms SLO; baseline: INFaaS on nonidentical EC2 setup. Within-SLO successes; comparator qualitative; not total throughput Source: printed p452 section6.1/Fig5; author PDF p10

Evidence ID	Metric and observation	Conditions and interpretation
E034	Clockwork request latency: approximately300 milliseconds	15 ResNet50 instances;16 clients/model;500ms SLO; baseline: No comparative ratio. Batch16 scheduling explanation; not a p99 value Source: printed p452 section6.1/Fig5; author PDF p10

A.10 S010 Colocating ML Inference and Training with Fast GPU Memory Handover (SIRIUS)

Report P010. Source: `https://www.usenix.org/system/files/atc25-wang-jiali.pdf`.

Quality assessment. 17/20 applicable points (85.0 percent).

Item	Score	Rationale	Evidence location
Q1	2	Inference/training colocation objective and scope explicit.	p1657 section1
Q2	2	VMM pool, memory handover and SM-mask control described.	pp1660-1665 sections3-5
Q3	1	Trace distributions and300s windows described; repeated-run uncertainty not established in checked evaluation.	pp1665-1666 section6
Q4	2	GPU, CPU/RAM, software and model sizes specified.	p1665 testbed; p1666 Table3
Q5	1	TaskSwitch is an adapted implementation; baseline choices explicit but implementation equivalence remains limited.	p1665 comparing targets
Q6	2	Training samples/s is defined; inference latency/SLO are kept distinct.	p1666 metrics/Figure10
Q7	2	P99, request arrivals, window and SLO definition supplied.	p1666 section6.1/Figure10
Q8	2	Co-located inference/training traces tested on one GPU.	p1666 section6.1/Figure10
Q9	1	Memory-reclamation boundary described; stronger fault/security validation not established.	pp1661-1665 sections4-5
Q10	2	Detailed implementation and evaluated artifact supplied; scope limitations discussed.	p1665 section5; p1669 section7; artifact appendix

Structured extraction.

Field	Source-supported content
Authors	Jiali Wang; Yankui Wang; Mingcong Han; Rong Chen
Year	2025
Venue	USENIX ATC2025, pp1657-1675
Mechanism	Temporal + MPS memory handover
Hardware and environment	V10016GB; Ubuntu20.04; CUDA11.6; PyTorch2.1.2; TVM0.14; LLM case A10080GB
Runtime details	Python3.10; CUDA11.6; PyTorch2.1.2; TVM0.14;
Workload	One GPU; six DNNs/up to56 instances + SWIN training; 300s; Poisson MAF; SLO4x solo
Tenants	six inference models duplicated round-robin to56 instances; co-located Swin-T training, batch72
Batch load and SLO	Light/Heavy/Burst/Skewed and compressed Azure Functions traces; 300s runs; SLO=4x standalone inference execution; Tidle5s, watermark1GB
Comparator	Reimplemented TaskSwitch; SP50/SP75; UM+MPS

Field	Source-supported content
Latency scope	P99 request latency and request-level SLO compliance; LLM TTFT/TBT evaluated separately
Throughput scope	training throughput in samples/s kept separate from inference metrics
Resource usage	explicit dynamic shared GPU-memory pool; model sizes from Table3; no invented GPU utilization percentage
Scalability	multi-GPU training evaluated separately in section6.2; excluded from local inference effect row
Reliability	Memory reclamation design is described;
Isolation boundary	resource handover and inference priority evaluated; no security/fault-containment guarantee inferred
Overhead	memory adjustment/allocation and batch discard evaluated in section6.3;
Advantages	supports separable inference latency objectives while opportunistically admitting training
Limitations	TaskSwitch is a reimplementation, not vanilla PipeSwitch.
Trade-offs	inference memory/SLO priority versus training throughput; hysteresis retains capacity to reduce thrashing
Source-linked finding	Inference p99/SLO outcomes accompany training progress; separate TTFT/TBT LLM case.

Evidence location: Section6.1 Fig10; section6.6.

Registered evidence rows.

Evidence ID	Metric and observation	Conditions and interpretation
E035	Average P99 inference latency improvement factor, as reported: 12.6 x	section6.1 single-V100 DNN inference+Swin-T; Light/Heavy/Burst/Skewed/MAF traces;300s; baseline: TaskSwitch (authors' adapted temporal-switch baseline). Author-reported average across tested workloads; not a uniform per-model speedup, and not a comparison to unmodified PipeSwitch. Source: printed p1666, section6.1/Figure10 discussion; PDF p11

A.11 S011 ParvaGPU: Efficient Spatial GPU Sharing for Large-Scale DNN Inference in Cloud Environments

Report P011. Source: `https://arxiv.org/pdf/2409.14447`.

Quality assessment. 14/20 applicable points (70.0 percent).

Item	Score	Rationale	Evidence location
Q1	2	Clear inference resource-sharing objective.	PDF pp1-2 introduction
Q2	2	MIG/MPS segments and algorithms explained.	PDF pp4-7 sectionIII
Q3	1	Profiling procedure and fleet scenarios described; repetition/uncertainty reporting unclear.	PDF pp5,8 sectionIV
Q4	1	Hardware/workload described; runtime version1.14.0 is printed but needs verification.	PDF p8 sectionIV-A
Q5	1	Local process comparison interpretable; fleet baselines adapted and no final-version equivalence established.	PDF pp5,8
Q6	2	Throughput axes specify requests/s and exact local values are given.	PDF pp4-5 Figure3
Q7	1	Latency milliseconds and settings stated; profiler quantile unclear.	PDF pp4-5 Figure4
Q8	2	Concurrent MPS process counts evaluated inside one MIG segment.	PDF pp4-5 Figures3-4
Q9	1	MIG boundary described; no direct security/fault validation established.	PDF pp2-4 background/design
Q10	1	Algorithms/settings detailed; accessed version remains preprint and replication verification incomplete.	PDF pp5-10

Structured extraction.

Field	Source-supported content
Authors	Munkyu Lee; Sihoon Seong; Minki Kang; Jihyuk Lee; Gap-Joo Na; In-Geol Chun; Dimitrios Nikolopoulos; Cheol-Ho Hong
Year	2024
Venue	SC2024 identity; extracted text is arXiv2409.14447v1
DOI	10.1109/SC41406.2024.00048
Mechanism	MIG + MPS
Hardware and environment	A10080GB; Ubuntu20.04; CUDA12.0.1; reported PyTorch1.14.0 needs checking
Runtime details	CUDA12.0.1; PyTorch1.14.0 as printed in source (unverified version string);
Workload	Eleven models; six scenario mixes; same-model MPS inside MIG
Tenants	one to three homogeneous MPS processes within a MIG instance; other models can occupy separate segments
Batch load and SLO	extracted example:1-GPC MIG instance; batch4;1/2/3 MPS processes; no tail quantile identified for local latency
Comparator	MIG-serving; gpulet and system variants
Latency scope	reported latency in ms; percentile not identified for profiler point

Field	Source-supported content
Throughput scope	requests/s; values354/444/446 at1/2/3 processes
Resource usage	MIG GPC allocation; no GPU utilization percentage inferred from number of GPUs used
Scalability	fleet allocator results remain separate from local measurement
Isolation boundary	MIG isolation is an architectural rationale; direct security/fault test not established
Overhead	scheduling/profiling discussed;
Advantages	joint choice of MIG size, MPS process count and batch size exposes throughput/latency trade-off
Limitations	Author manuscript; unusual runtime version; GPU count is not occupancy.
Trade-offs	additional MPS processes improve local throughput modestly while increasing latency under saturated small MIG instances
Source-linked finding	Fig7 separates internal/external fragmentation; Fig5 counts GPUs.
Report version	Peer-reviewed SC24 identity confirmed by official proceedings. Detailed extraction uses the linked author manuscript; final-publication value equivalence is a source-version limitation.

Evidence location: PDF p8 onward; sectionIV; TableIV; Figs5/7.

Registered evidence rows.

Evidence ID	Metric and observation	Conditions and interpretation
E036	Reported throughput: 446 requests/s	InceptionV3,1-GPC MIG instance,batch4;3 MPS processes; baseline: same configuration with1 process (354requests/s). One local configuration, not fleet savings. Source is arXivv1; final published version not verified. Source: author preprint PDF p5, sectionIII-B prose; Figure3 p4

Evidence ID	Metric and observation	Conditions and interpretation
E037	Reported inference latency; quantile unspecified: 27 ms	InceptionV3,1-GPC MIG instance,batch4;3 MPS processes; baseline: same configuration with1 process (11ms). Quantile/arrival model not established for this profiler point; do not label p99. Value is specific to the accessed author manuscript. Source: author preprint PDF p5, sectionIII-B prose; Figure4 p4

A.12 S012 Torpor: GPU-Enabled Serverless Computing for Low-Latency, Resource-Efficient Inference

Report P012. Source: `https://www.usenix.org/system/files/atc25-yu.pdf`.

Quality assessment. 15/20 applicable points (75.0 percent).

Item	Score	Rationale	Evidence location
Q1	2	Inference memory/density objective explicit.	p597 section1
Q2	2	API redirection, pooling and scheduling described.	pp600-604 sections4-6
Q3	1	Production trace and request rates given; repeated-run uncertainty not established.	pp604-606 section7
Q4	1	GPU/CPU/RAM supplied; exact software version matrix missing in checked setup.	p604 settings
Q5	1	Native and adapted INFless-KA comparison described; resource/CPU paths differ.	pp605-606 sections7.1-7.3
Q6	2	Throughput/load definitions and local normalization supplied.	pp605-606 Figure8
Q7	2	P98 and per-function load/SLO explicit.	pp605-606 metrics/Figure8
Q8	2	Multiple functions share a single GPU in Figure8.	pp605-606 Figure8
Q9	1	Sharing/isolation modes separated; strong security/fault tests not established for evaluated mode.	sections4,7-8
Q10	1	Implementation detail and limitations supplied; independent artifact reproduction unperformed.	sections6-8

Structured extraction.

Field	Source-supported content
Authors	Minchen Yu; Ao Wang; Dong Chen; Haoxuan Yu; Xiaonan Luo; Zhuohao Li; Wei Wang; Ruichuan Chen; Dapeng Nie; Haoran Yang; Yu Ding
Year	2025
Venue	USENIX ATC2025 pp597-612
Mechanism	API remoting + temporal swapping
Hardware and environment	Worker48vCPU/384GB; four V10032GB
Runtime details	PyTorch-based clients replace CUDA libraries; exact OS/driver/CUDA versions not reported in section7 settings
Workload	Eight inference models; warm-up; CV80ms/BERT200ms SLO
Tenants	multiple low-frequency ResNet152 functions share one GPU; sufficient functions to saturate device memory
Batch load and SLO	10-80 requests/minute/function; global defaultP98; CV deadline80ms and Bert-qa200ms
Comparator	Native; remote GPU; PCIe/NVLink swap
Latency scope	P98 under shared-GPU Figure8; Table4 isolated latency has no explicitly stated quantile
Throughput scope	single-GPU throughput normalized to Native, request rates varied

Field	Source-supported content
Resource usage	GPU load=time processing inference / observation time; host memory retains unloaded models
Scalability	up to6 workers/4GPUs per worker evaluated separately; local outcome rows do not claim fleet effects
Energy	not reported as measured outcome in checked sections7-8
Isolation boundary	runtime sharing and runtime isolation modes differ; section7 measures sharing mode, not security guarantee
Overhead	API redirection, swapping and extra CPU activity evaluated; ResNet152 remoting17ms versus Native25ms, Table4
Advantages	more low-rate functions can share GPU memory capacity through late binding
Limitations	p98 tail; busy time is not SM occupancy; node scaling separate.
Trade-offs	device-memory density versus transfer traffic, CPU work and tail latency
Source-linked finding	BERT Table4: native42ms, remote43ms, PCIe swap144ms, NVLink swap45ms.

Evidence location: PDF pp10-11; Table4; sections7.2-7.3.

Registered evidence rows.

Evidence ID	Metric and observation	Conditions and interpretation
E038	Reported single-GPU throughput factor: >10 x	multiple ResNet152 functions on one V10032GB;10requests/minute/function; baseline: Native early binding at same offered function rate. Lower bound reported by authors, not an exact10x; function count varies to saturate memory. No fleet cost claim. Source: p606 section7.2; Figure8 p605; PDF pp10-11

Evidence ID	Metric and observation	Conditions and interpretation
E039	Reported P98 latency upper bound: <50 ms	same single-GPU low-rate ResNet152 test,10-80requests/minute/function; baseline: Native under matched request-rate setting. Bound from text, not a digitized point; default98th percentile from section7 metrics. Applies to this workload. Source: p606 section7.2; Figure8 p605; PDF pp10-11

A.13 S013 StreamBox: A Lightweight GPU SandBox for Serverless Inference Workflow

Report P013. Source: `https://www.usenix.org/system/files/atc24-wu-hao.pdf`.

Quality assessment. 15/20 applicable points (75.0 percent).

Item	Score	Rationale	Evidence location
Q1	2	Workflow-local runtime sharing objective explicit.	p59 section1
Q2	2	Streams,PTB compiler and memory management explained.	pp62-67 sections3-8
Q3	1	Production-style traces and ablations provided; repeated-run uncertainty unclear.	pp67-70 section9
Q4	1	Hardware/frameworks named; OS/driver and version compatibility need checking.	p67 setup
Q5	1	Common optimizations enabled, but baselines adapted and batch choices affect ratios.	p67 Table2/comparing targets
Q6	2	Memory and throughput units/ratios defined.	p68 Figures12-13
Q7	1	CDF/SLO reported; no precise tail scalar extracted.	p68 Figure14
Q8	2	Explicit3/6-user single-GPU test.	p70 Figure19
Q9	1	Same-runtime sharing weakens function isolation; security boundary is deliberately scoped.	pp62-63; p70 related-work boundary
Q10	2	Detailed implementation, preparation overhead and multi-user limits supplied.	p70 sections9.3-9.4

Structured extraction.

Field	Source-supported content
Authors	Hao Wu; Yue Yu; Junxiao Deng; Shadi Ibrahim; Song Wu; Hao Fan; Ziyue Cheng; Hai Jin
Year	2024
Venue	USENIX ATC2024 pp59-73
Mechanism	API/runtime sandbox virtualization
Hardware and environment	Four V10016GB; 28cores/128GB; CUDA10.1; PyTorch1.3; TensorFlow2.12
Runtime details	PyTorch1.3.0,TensorFlow2.12,CUDA10.1 as reported; driver/OS version not reported in section9 setup
Workload	Three concurrent inference workflows/GPU
Tenants	concurrent functions within workflow share runtime; separate3/6-user Traffic test on one GPU
Batch load and SLO	Azure Functions7-day invocation trace; sporadic/periodic/bursty; Traffic SLO200ms; up to3 workflows/runtime contexts per GPU in evaluated setup
Comparator	MPS; INFless; Astraea; Stream-only
Latency scope	end-to-end CDF and SLO violations; no invented p99 scalar
Throughput scope	average requests/s and within-experiment ratios
Resource usage	runtime memory redundancy, automatic pool sizing and direct intra-GPU data sharing

Field	Source-supported content
Scalability	single-GPU3/6-user limits versus separate4/8-node scaling tests
Energy	not reported as measured outcome in checked section9
Isolation boundary	functions within one workflow share CUDA address space; secure cross-workflow isolation is orthogonal; do not claim strong per-function isolation
Overhead	API forwarding1-3microseconds;3-5microseconds at48functions; one-time profiling36s and transformation4s, p70
Advantages	reduces redundant runtime memory and inter-function data copies
Limitations	Sandbox memory separation is not proof of adversarial security.
Trade-offs	workflow-local sharing and compiler/profiling preparation versus runtime density and startup latency
Source-linked finding	Table2 evaluates runtime sandbox workflows on shared GPUs.

Evidence location: PDF p10 onward; section9; Table2.

Registered evidence rows.

Evidence ID	Metric and observation	Conditions and interpretation
E040	Author-reported average throughput factor: 6.7 x	three workflow types,PyTorch/TensorFlow,Azure-derived invocation patterns; evaluated V100 setup; baseline: INFless with documented warm-up/pipelined loading/model sharing. Aggregate within-paper result across tested workflows; not universal or a matched cross-paper speedup. Batch/concurrency optimizations contribute. Source: p68 section9.1/Figure13; PDF p11

Evidence ID	Metric and observation	Conditions and interpretation
E041	Reported API-forwarding overhead range: 3-5 microseconds	GPU API forwarding with48concurrent functions on one GPU; baseline: direct API path; instrumentation overhead measurement. Per-API overhead, not end-to-end inference latency; many APIs asynchronous. Source: p70 section9.3/Figure18; PDF p13

A.14 S014 Efficient Performance-Aware GPU Sharing with Compatibility and Isolation through Kernel Space Interception (KRYPTON)

Report P014. Source: `https://www.usenix.org/system/files/atc25-zhang-shulai.pdf`.

Quality assessment. 16/20 applicable points (80.0 percent).

Item	Score	Rationale	Evidence location
Q1	2	Sharing/isolation/compatibility objectives explicit.	p1003 section1
Q2	2	Kernel interception,token control and MIG orchestration described.	pp1006-1011 sections3-4
Q3	1	Setup and workload matrix supplied; repeated-run uncertainty unclear.	p1011 section5.1
Q4	2	Hardware,OS,kernel,driver and frameworks specified.	p1011 Tables1-2
Q5	1	Isolation-capable baselines and adapters disclosed; equivalence limited.	pp1011-1012 comparing targets
Q6	2	Quota/throughput metrics explicitly normalized.	pp1013-1014 sections5.4-5.6
Q7	2	QoS uses originalP99 and defined Poisson load/token lengths.	p1014 Figure17/section5.5
Q8	2	Local co-running inference error measured.	p1014 section5.6
Q9	1	Boundary/security rationale described; direct fault/security validation not established in checked evaluation.	p1015 section7
Q10	1	Implementation andMIGlimitations supplied; independent artifact reproduction unperformed.	pp1011,1015 sections4,7

Structured extraction.

Field	Source-supported content
Authors	Shulai Zhang; Ao Xu; Quan Chen; Han Zhao; Weihao Cui; Zhen Wang; Yan Li; Limin Xiao; Minyi Guo
Year	2025
Venue	USENIX ATC2025 pp1003-1019
Mechanism	Kernel API virtualization + hybrid
Hardware and environment	A10040GB/RTX4090; Ubuntu20.04; kernel5.15.0-91; driver550.90.07; CUDA12.1; PyTorch2.2.1
Runtime details	NVIDIA550.90.07;Docker24.0.7;CUDA12.1;Vulkan1.3;PyTorch2.2.1
Workload	Inference/training/rendering in separate containers; solo-throughput targets
Tenants	inference with co-running applications; separate training/rendering results not substituted
Batch load and SLO	QoS:Poisson arrivals at half maximum throughput;50%temporal allocation;5/10/20/40/80ms tokens; target original soloP99
Comparator	Temporal; MIG; adapted gpulet

Field	Source-supported content
Latency scope	fraction of requests exceeding originalP99 threshold
Throughput scope	local normalized-throughput target errors separated from total GPU-count savings
Resource usage	4.8MB host memory for kernel module;one CPU monitoring thread;NVML/DCGM monitoring
Scalability	fleet packing evaluated separately; no local32.1%GPU-count claim
Energy	not reported as measured outcome in checked section5
Reliability	production reliability claimed in discussion; not independent fault-testing evidence
Isolation boundary	separate virtual-device/application boundary; kernel module trust remains; security discussion is not proof of no vulnerabilities
Overhead	token-length trade-off and4.8MBhost-memory module overhead
Advantages	multi-runtime support and feedback-controlled quotas without application rewrites
Limitations	Baseline adaptations and workload mix constrain attribution.
Trade-offs	shorter scheduling tokens increase overhead; longer tokens increase waiting for larger batches
Source-linked finding	Inference-specific sharing comparisons; section2.2 tests fault propagation.

Evidence location: PDF p10 Tables1-2; section5; section2.2 Fig3.

Registered evidence rows.

Evidence ID	Metric and observation	Conditions and interpretation
E042	Reported average QoS violation rate: 1.23 percent of requests	Transformer-XL inference,batch8/16/32;50%GPUtime;Poisson arrivals;tokens5-80ms; baseline: original soloP99 threshold (1%baseline violations by definition). Average across tested configurations; no claim that worst-case violation is1.23%. Threshold-relative metric, not latency in milliseconds. Source: p1014 section5.5/Figure17; PDF p13

Evidence ID	Metric and observation	Conditions and interpretation
E043	Reported kernel module host-memory footprint: 4.8 MB	kernel-space GPU sharing module on evaluated Linux host; baseline: no module; additional host memory cost. Host memory, not GPU VRAM; separate monitoring thread also required. Source: p1014 section5.7; PDF p13

A.15 S015 XSched: Preemptive Scheduling for Diverse XPUs

Report P015. Source: `https://www.usenix.org/system/files/osdi25-shen-weihang.pdf`.

Quality assessment. 16/18 applicable points (88.9 percent).

Item	Score	Rationale	Evidence location
Q1	2	Portable scheduling objective andGPUcase clear.	pp671-672 section1
Q2	2	XQueue abstraction andmulti-level preemption described.	pp674-679 sections3-6
Q3	1	Workload rates and policies explicit; repeat uncertainty not established.	pp680-683 section7
Q4	1	GPU/model/rates explicit;host/software versions incomplete in extracted case.	p683 case3
Q5	2	Same-GPU standalone/native priority comparisons supplied.	p683 Figure15
Q6	2	Throughput-latency curves defined forGPUinferencecase.	p683 Figure15b
Q7	2	P99 and offered rates explicit.	p683 Figure15a
Q8	2	Two concurrentBertclients ononeGV100.	p683 case3
Q9	NA	No claimed security/fault-containment guarantee in extracted scheduling claim;N/A with stated rationale for claim-specificQ9.	pp671-672 scope; p683 case3
Q10	2	Implementation/interfaces,artifact andhardware-dependence detailed.	sections6-8;artifact appendix

Structured extraction.

Field	Source-supported content
Authors	Weihang Shen; Mingcong Han; Jialong Liu; Rong Chen; Haibo Chen
Year	2025
Venue	USENIX OSDI2025 pp671-692
Mechanism	Temporal API interposition
Hardware and environment	One GV100 inference case; broader XPU experiments separate
Runtime details	CUDA/Triton and Paella artifact; exact version numbers not reported in extracted case3
Workload	Two BERT-large clients; HP10req/s; saturated BE
Tenants	one high-priority client at10requests/s;low-priority client continuous on sameGV100
Batch load and SLO	Triton client rates as above;Paella workload lognormal sigma2.0;K-EDF K16; exact batch sizes not reported in extracted case3
Comparator	Triton; Triton+priority; Triton+XSched
Latency scope	high-priorityP99 normalized to standalone;latency-throughput curve
Throughput scope	requests/s in Paella curve;no aggregate cross-XPU overhead used as GPU inference result
Resource usage	preemption enables foreground service priority; no invented GPU utilization percentage
Scalability	ten-XPU portability evaluation distinguished from one-GPU case

Field	Source-supported content
Energy	not reported as measured outcome in checked section7
Isolation boundary	scheduling/performance priority, not a security or fault-containment guarantee
Overhead	10lines Triton integration and15lines Paella-baseline integration;preemption overhead measured separately
Advantages	adds priority policy to existing serving frameworks with small integration changes
Limitations	Data assumed to fit memory; CUDA graphs and undocumented driver limits.
Trade-offs	foreground predictability versus background throughput and hardware-specific adaptation
Source-linked finding	Fig15: relative p99 1.53/1.51/1.07 respectively versus solo.

Evidence location: PDF pp14-15; section8 Case3 Fig15; section9.

Registered evidence rows.

Evidence ID	Metric and observation	Conditions and interpretation
E044	Reported high-priorityP99 latency ratio: 1.07 x standalone	twoBert-large clients ononeGV100;high-priority10req/s,backgroundcontinuous; baseline: standalone high-priorityBert-large execution. Triton+XSched;not a rawmillisecond value. NativeTriton ratio1.53x;noNPUresultincluded. Source: p683 section7.4 case3/Figure15a; PDF p14

Evidence ID	Metric and observation	Conditions and interpretation
E045	Author-reported P99 latency reduction: 30.0 percent	same two-clientTriton/GV100case; baseline: vanillaTriton (P99=1.53xstandalone). Within-paper comparison only;author-rounded30.0%,not a universal bound. Source: p683 section7.4 case3/Figure15a; PDF p14

A.16 S016 InferFair: Towards QoS-aware scheduling for performance isolation guarantee in heterogeneous model serving systems

Report P018. Source: `https://www.sciencedirect.com/science/article/pii/S0167739X23003187`.

Quality assessment. 15/20 applicable points (75.0 percent).

Item	Score	Rationale	Evidence location
Q1	2	Heterogeneous inference fairness is explicit	Published pp17-18; sections5.1-5.4; Table3; Eq2; Markdown lines718-899
Q2	2	Temporal scheduler and feedback controller are described	Published pp17-18; sections5.1-5.4; Table3; Eq2; Markdown lines718-899
Q3	1	Configuration/control details are incomplete in accessible extraction	Published pp17-18; sections5.1-5.4; Table3; Eq2; Markdown lines718-899
Q4	2	Models, arrival rates and SLOs are explicit	Published pp17-18; sections5.1-5.4; Table3; Eq2; Markdown lines718-899
Q5	1	EDF lacks batching and is not a matched baseline	Published pp17-18; sections5.1-5.4; Table3; Eq2; Markdown lines718-899
Q6	2	Goodput and unfairness retain separate definitions	Published pp17-18; sections5.1-5.4; Table3; Eq2; Markdown lines718-899
Q7	1	Service SLOs are given; request-tail quantile evidence is limited	Published pp17-18; sections5.1-5.4; Table3; Eq2; Markdown lines718-899
Q8	2	Multiple concurrent inference models are evaluated	Published pp17-18; sections5.1-5.4; Table3; Eq2; Markdown lines718-899
Q9	1	Performance fairness does not establish security/fault separation	Published pp17-18; sections5.1-5.4; Table3; Eq2; Markdown lines718-899
Q10	1	Algorithms and limits are interpretable; software-version coverage is incomplete	Published pp17-18; sections5.1-5.4; Table3; Eq2; Markdown lines718-899

Structured extraction.

Field	Source-supported content
Authors	Yaqiong Peng and Haocheng Peng
Year	2024
Venue	Future Generation Computer Systems
DOI	10.1016/j.future.2023.08.020
Mechanism	Temporal SLO-aware scheduling + feedback goodput balancing; MPS comparator

Field	Source-supported content
Hardware and environment	Dell PowerEdgeR740; V10032GB; local synthetic co-running experiment and separate four-GPU realistic trace.
Workload	Local GoogLeNet/ResNet152 Poisson arrivals; three-model test adds Mobile_Pose_MobileNetv3;600req/s distributed in six ratios; SLO100ms and interval50ms. Separate61-model,90min MAF replay on four GPUs.
Comparator	Clockwork; MPS plus Clockwork SLO scheduling with two worker processes/GPU; EDF without batching (not a like-for-like batching baseline).
Limitations	Two-column Markdown interleaves algorithms/prose; use labelled Table3 rows and sections, not scrambled formulas. Repetitions/variance and software versions unresolved. MAF trace has minute-level invocation counts; cross-study batching and isolation definitions differ.
Source-linked finding	Local Table3 load3:1:2: unfairness1.05 vs Clockwork1.50; SLO goodput524.29 vs457.24req/s. Other load ratios give smaller differences. Local unfairness is max/min model slowdown(Eq2), not p99 or security. Four-GPU MAF outcome kept separate.
Performance isolation	Relative goodput/slowdown fairness and SLO-compliant goodput; no extracted p99 bound
Security boundary	Not established by performance-isolation terminology
Fault boundary	Not tested in extracted results

Evidence location: Published pp17-18; sections5.1-5.4; Table3; Eq2; Markdown lines718-899.

Registered evidence rows.

Evidence ID	Metric and observation	Conditions and interpretation
E046	SLO-compliant goodput: 524.29 requests/s	Local three-model load ratio 3:1:2; baseline: Clockwork: 457.24 requests/s. Local outcome, not four-GPU replay or p99 Source: Published pp. 17-18; Table 3

Evidence ID	Metric and observation	Conditions and interpretation
E047	Unfairness ratio: 1.05 ratio	Local three-model load ratio 3:1:2; baseline: Clockwork: 1.50. Maximum/minimum model slowdown; not security isolation Source: Published pp. 17-18; Table 3; Eq. 2

A.17 S017 ArkGPU: enabling applications' high-goodput co-location execution on multitasking GPUs

Report P019. Source: `https://doi.org/10.1007/s42514-023-00154-y`.

Quality assessment. 16/20 applicable points (80.0 percent).

Item	Score	Rationale	Evidence location
Q1	2	Latency-critical inference under co-location is explicit	Published pp313-319; sections4.3,5.1-5.4; Table3; Figs14-21; Markdown lines920-1360
Q2	2	Interception, NVML and throttling are specified	Published pp313-319; sections4.3,5.1-5.4; Table3; Figs14-21; Markdown lines920-1360
Q3	1	Repeated-run precision is not established by the available source assessment	Published pp313-319; sections4.3,5.1-5.4; Table3; Figs14-21; Markdown lines920-1360
Q4	2	GPU, software and thirty model pairs are described	Published pp313-319; sections4.3,5.1-5.4; Table3; Figs14-21; Markdown lines920-1360
Q5	1	Exact comparator MPS quotas are not established	Published pp313-319; sections4.3,5.1-5.4; Table3; Figs14-21; Markdown lines920-1360
Q6	2	QPS and goodput are distinguished	Published pp313-319; sections4.3,5.1-5.4; Table3; Figs14-21; Markdown lines920-1360
Q7	2	p99 criterion is twice solo latency	Published pp313-319; sections4.3,5.1-5.4; Table3; Figs14-21; Markdown lines920-1360
Q8	2	Thirty inference/training pairs are tested	Published pp313-319; sections4.3,5.1-5.4; Table3; Figs14-21; Markdown lines920-1360
Q9	1	No adversarial or cross-tenant fault test is established	Published pp313-319; sections4.3,5.1-5.4; Table3; Figs14-21; Markdown lines920-1360
Q10	1	Overhead and low-contention regression are discussed; replication gaps remain	Published pp313-319; sections4.3,5.1-5.4; Table3; Figs14-21; Markdown lines920-1360

Structured extraction.

Field	Source-supported content
Authors	Jie Lou and Yiming Sun and Jie Zhang and Huawei Cao and Yuan Zhang and Ninghui Sun
Year	2023
Venue	CCF Transactions on High Performance Computing
DOI	10.1007/s42514-023-00154-y
Mechanism	CUDA API interception + NVML monitoring and admission throttling
Hardware and environment	Machine with four V100s; Xeon Platinum8163; CentOS7.9.2009; driver460.106.00; CUDA11.1; cuDNN7.6.5.
Workload	Five MLPerf inference LC models (ResNet50, SSD-MobileNet, SSD-ResNet34, DLRM, BERT), each paired locally with six BE training models (30 pairs). Single-stream query size1; p99 QoS threshold twice solo. Separate four-GPU app-mixes are not local occupancy evidence.
Comparator	MPS (Non-ArkGPU); k8s-native for QPS/utilization; GaiaGPU and adjustment-unit ablation.
Limitations	Mixed inference/training qualifies via explicit inference p99/QPS and workload transfer rationale; it does not establish inference-only performance. Polling/API sleeps can add delay; BE checkpoint/restart affects costs. Huge goodput ratios can have near-zero baseline yield. Repetitions/variance and chart coordinates unresolved.
Source-linked finding	Section5.2 reports all30 tested pairs satisfy the 2x-solo p99 QoS target with ArkGPU. Low-contention pairs can lose about5% QPS from throttling. Four-GPU app-mix goodput gains94.98% vs k8s-native and38.65% vs MPS are separate system-level results.
Performance isolation	p99 tested against twice-solo QoS; average latency, QPS and goodput separately
Security boundary	Not established by API interception
Fault boundary	Background discussion of MPS faults is not an ArkGPU fault test

Evidence location: Published pp313-319; sections4.3,5.1-5.4; Table3; Figs14-21; Markdown lines920-1360.

Registered evidence rows.

Evidence ID	Metric and observation	Conditions and interpretation
E048	Pairs meeting twice-solo p99 QoS target: 30 tested pairs	Five inference models paired with six training models; baseline: MPS Non-ArkGPU. Count of tested pairs, not a universal guarantee Source: Published pp. 313-319; Section 5.2

Evidence ID	Metric and observation	Conditions and interpretation
E049	Low-contention QPS loss: about 5 percent	Low-contention tested pair settings; baseline: Unthrottled co-location. Regression retained; larger system-level goodput gains are not one-GPU results Source: Section 5.2 source assessment

A.18 S018 Dyna-P: placement-aware dynamic partitioning for lightweight applications with modern GPUs

Report P020. Source: `https://doi.org/10.1007/s10586-025-05284-2`.

Quality assessment. 15/20 applicable points (75.0 percent).

Item	Score	Rationale	Evidence location
Q1	2	Fragmentation and concurrent inference are explicit	PDF pp11-16; sections4.1-4.3.2; Table2; Figs8-13; Markdown lines700-865
Q2	2	MIG placement and merge/split policies are described	PDF pp11-16; sections4.1-4.3.2; Table2; Figs8-13; Markdown lines700-865
Q3	1	Repetition/variance detail is limited in accessible evidence	PDF pp11-16; sections4.1-4.3.2; Table2; Figs8-13; Markdown lines700-865
Q4	2	GPU profiles and local inference job ranges are specified	PDF pp11-16; sections4.1-4.3.2; Table2; Figs8-13; Markdown lines700-865
Q5	1	MISO adaptation and privileged ORACLE require caveats	PDF pp11-16; sections4.1-4.3.2; Table2; Figs8-13; Markdown lines700-865
Q6	2	Makespan and STP are separate metrics	PDF pp11-16; sections4.1-4.3.2; Table2; Figs8-13; Markdown lines700-865
Q7	1	JCT is not online request p99	PDF pp11-16; sections4.1-4.3.2; Table2; Figs8-13; Markdown lines700-865
Q8	2	Local concurrent inference job strata are identifiable	PDF pp11-16; sections4.1-4.3.2; Table2; Figs8-13; Markdown lines700-865
Q9	1	Architectural MIG descriptions do not constitute adversarial tests	PDF pp11-16; sections4.1-4.3.2; Table2; Figs8-13; Markdown lines700-865
Q10	1	Reconfiguration and prediction limits are explicit; software coverage is incomplete	PDF pp11-16; sections4.1-4.3.2; Table2; Figs8-13; Markdown lines700-865

Structured extraction.

Field	Source-supported content
Authors	Theodora Adufu and Yoonhee Kim
Year	2025
Venue	Cluster Computing
DOI	10.1007/s10586-025-05284-2
Mechanism	MIG dynamic placement/reconfiguration with reconfiguration-last policy
Hardware and environment	Two A30s,24GB each; CUDA12.2; driver535.171.04; DCGM3.1.3. Local sections4.2/4.3.1 use one A30. StaCon profiles2g.12gb+1g.6gb+1g.6gb.

Field	Source-supported content
Workload	50 SLM inference jobs; batch1-64; DistilGPT, Codegen, Flan-T5-Large, Codellama and Phi-1 as labelled in Table2. Local batch sensitivity:10 inference jobs, batches1-8 vs32-64 on one A30. Later160-A100 cluster study is simulation.
Comparator	Full-GPU sequential baseline; fixed StaCon MIG; pre-profiled MISO adapted to A30; ORACLE with future job/size/time knowledge.
Limitations	Retain one-GPU inference strata separately from two-GPU aggregate and cluster simulation. JCT/makespan/STP cannot replace request-tail latency or goodput. Reconfiguration can stop/restart active jobs; neither architectural MIG claims nor OOM avoidance prove hostile isolation. Model labels/parameter counts and corrupted overhead fraction require caution.
Source-linked finding	Single-A30 batch experiments: smaller partitions increase concurrency but can increase mean JCT; not online p99 evidence. Fig8 broader job set reports39.03% lower makespan vs StaCon,12.14% vs MISO and14.7% higher normalized STP vs StaCon. Prediction-error sensitivity:15% perturbation adds8.09% makespan.
Performance isolation	Job completion/makespan/STP and prediction sensitivity; no extracted online p99 guarantee
Security boundary	Architectural isolation discussed, no adversarial test
Fault boundary	OOM observation is not cross-tenant fault containment validation

Evidence location: PDF pp11-16; sections4.1-4.3.2; Table2; Figs8-13; Markdown lines700-865.

Registered evidence rows.

Evidence ID	Metric and observation	Conditions and interpretation
E050	Effect of 15 percent prediction perturbation: 8.09 percent makespan increase	Reported prediction sensitivity; baseline: Unperturbed predictor. Sensitivity result; local JCT and cluster simulation are separate Source: PDF pp. 11-16; Section 4.3.2

A.19 S019 KRISP: Enabling Kernel-wise RIght-sizing for Spatial Partitioned GPU Inference Servers

Report P021. Source: `https://danielwong.org/files/KRISP-HPCA2023.pdf`.

Quality assessment. 16/20 applicable points (80.0 percent).

Item	Score	Rationale	Evidence location
Q1	2	Kernel-level right-sizing objective is explicit	PDF pp9-10; sectionsV-VI; TableIII
Q2	2	API control and spatial sizing are classifiable	PDF pp9-10; sectionsV-VI; TableIII
Q3	1	Feature emulation limits direct experimental transfer	PDF pp9-10; sectionsV-VI; TableIII
Q4	2	MI50 and tested inference workers/batches are specified	PDF pp9-10; sectionsV-VI; TableIII
Q5	1	MPSDefault is AMD unrestricted concurrency, not NVIDIA MPS	PDF pp9-10; sectionsV-VI; TableIII
Q6	2	Throughput is defined for the tested worker configuration	PDF pp9-10; sectionsV-VI; TableIII
Q7	2	p95 inference latency is evaluated	PDF pp9-10; sectionsV-VI; TableIII
Q8	2	Concurrent inference workers and heterogeneous pairs are tested	PDF pp9-10; sectionsV-VI; TableIII
Q9	1	Resource sizing is not a validated security/fault boundary	PDF pp9-10; sectionsV-VI; TableIII
Q10	1	Emulation and subtracted overhead limit deployability	PDF pp9-10; sectionsV-VI; TableIII

Structured extraction.

Field	Source-supported content
Authors	Marcus Chow and Ali Jahanshahi and Daniel Wong
Year	2023
Venue	IEEE HPCA
DOI	10.1109/HPCA56546.2023.10071121
Mechanism	Spatial partition/API control (AMD)
Hardware and environment	One MI50/60CUs; dualEPYC7302/512GB; Ubuntu18.04; kernel5.4; ROCm4.5
Workload	One/two/four workers and heterogeneous pairs; batch32; aggregate8/16; eight models
Comparator	AMD unrestricted concurrency labelled MPSDefault
Limitations	Feature emulated; overhead subtracted; baseline is not NVIDIA MPS.
Source-linked finding	Kernel-wise right-sizing evaluates throughput and p95 inference latency.

Evidence location: PDF pp9-10; sectionsV-VI; TableIII.

Registered evidence rows.

Evidence ID	Metric and observation	Conditions and interpretation
E051	Evaluated latency statistic: 95 percentile	Inference workers and heterogeneous pairs; baseline: AMD unrestricted concurrency labelled MPSDefault. Statistic definition, not a 95-percent latency improvement Source: PDF pp. 9-10; Sections V-VI; Table III

A.20 S020 Tally: Non-Intrusive Performance Isolation for Concurrent Deep Learning Workloads

Report P022. Source: `https://arxiv.org/pdf/2410.07381`.

Quality assessment. 17/20 applicable points (85.0 percent).

Item	Score	Rationale	Evidence location
Q1	2	Performance isolation during co-location is explicit	PDF pp9-11; sections5-5.3; Fig6
Q2	2	CUDA interception and priority control are described	PDF pp9-11; sections5-5.3; Fig6
Q3	1	Repeated-run precision is not established in the checked source summary	PDF pp9-11; sections5-5.3; Fig6
Q4	2	GPU and principal software versions are specified	PDF pp9-11; sections5-5.3; Fig6
Q5	1	TGS and scheduling policies trade different background yields	PDF pp9-11; sections5-5.3; Fig6
Q6	2	Foreground protection and background throughput remain distinct	PDF pp9-11; sections5-5.3; Fig6
Q7	2	High-priority p99 is an evaluated outcome	PDF pp9-11; sections5-5.3; Fig6
Q8	2	Same-GPU co-tenant pairs and production-style traces are used	PDF pp9-11; sections5-5.3; Fig6
Q9	1	Priority control does not establish hostile memory/fault isolation	PDF pp9-11; sections5-5.3; Fig6
Q10	2	Artifact, implementation and throughput sacrifice support interpretation	PDF pp9-11; sections5-5.3; Fig6

Structured extraction.

Field	Source-supported content
Authors	Wei Zhao and Anand Jayarajan and Gennady Pekhimenko
Year	2025
Venue	ACM ASPLOS Volume 1
DOI	10.1145/3669940.3707282
Mechanism	CUDA API interception + priority
Hardware and environment	A10040GB; Ubuntu20.04; CUDA12.2; PyTorch2.2; ONNX1.17
Workload	Same-GPU pairs; MAF inference traces; high-priority p99
Comparator	Time slicing; MPS; priority MPS; TGS
Limitations	Latency protection sacrifices some BE throughput; author manuscript.
Source-linked finding	At 10% idle time: normalized throughput Tally1.19 vs TGS1.65.

Evidence location: PDF pp9-11; sections5-5.3; Fig6.

Registered evidence rows.

Evidence ID	Metric and observation	Conditions and interpretation
E052	Normalized background throughput: 1.19 ratio	Ten percent idle-time setting; baseline: TGS: 1.65. Foreground protection trades background throughput; compare this setting only Source: PDF pp. 9-11; Sections 5-5.3; Figure 6

B Calculated quantities

ID	Formula and inputs	Calculated value and scope
C001	(446/354-1)*100; 446requests/s at3processes;354requests/s at1process;InceptionV3,1-GPC,batch4	25.988700564971758 percent; ParvaGPU arXivv1 PDF p5 sectionIII-B/Figure3
C002	27/11; 27ms at3processes;11ms at1process;same1-GPC/batch4configuration	2.4545454545454546 x; ParvaGPU arXivv1 PDF p5 sectionIII-B/Figure4

C Revised protocol amendment

Amendment B007 records the user-authorized deadline continuation on 17 October 2026. The report uses one consolidated, software-supported screening and appraisal process and a bounded twenty-study comparative synthesis. The original independent-review protocol remains preserved. Source assessments retain their evidence locations, and external scoring scales are not substituted for the submitted Q1–Q10 rubric. Records outside the stopping rule remain visible in the accounting. This amendment documents the implemented review method and coverage boundary.

The data archive includes the original baseline, the revised screening decisions, two source exclusions, 200 quality-item appraisals, twenty extraction forms, 52 evidence rows and two calculations. Stable identifiers preserve every correction and original scope note.

D Submitted protocol



NED UNIVERSITY OF ENGINEERING AND TECHNOLOGY

Department of Software Engineering

Course: OPERATING SYSTEMS

Course code: SE-303 Instructor: Dr. Mustafa Latif Section: A

SYSTEMATIC LITERATURE REVIEW (SLR) PROTOCOL

GPU Sharing Mechanisms for Multi-Tenant Inference Serving: Time-Slicing vs. MPS vs. MIG vs. API Remoting

A Study of Utilization, Latency, and Isolation in Operating Systems

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27 September 2026

1 Background and Motivation

Modern machine-learning inference systems may serve multiple applications, models or customers on a shared GPU. In this protocol, a *tenant* is a separately managed inference workload or process whose performance may be affected by co-tenants. A single physical GPU may be shared to improve use of capacity and support more tenants; the magnitude of any benefit depends on workload and configuration and is an empirical question. GPU resource management involves driver/runtime and device-level behavior as well as operating-system scheduling. Hardware support for preemption and isolation varies by GPU and software stack; it must not be generalized from one product generation to all GPUs.

Table 1 distinguishes the four assigned comparison families from related implementation labels. NVIDIA's documentation defines MPS as a service allowing multiple CUDA processes to share GPU resources, MIG as partitioning supported GPUs into instances, and time-slicing as a form of shared access with weaker memory/fault isolation than MIG in its GPU Operator setting [1, 2, 3]. Vendor documentation establishes terminology and documented capabilities, not independent comparative inference results.

Table 1: Mechanism framework and scope

Mechanism	Basic approach	Review role
Time-slicing	Multiple tenants obtain GPU execution over different time intervals; configuration and actual preemption behavior are recorded.	Core temporal family
NVIDIA MPS	CUDA processes share GPU execution resources through the MPS service; exact limits/settings depend on version.	Core MPS family
NVIDIA MIG	A supported physical GPU is partitioned into hardware GPU instances with documented isolation properties and fixed profiles.	Core spatial family
GPU virtualization	Virtual GPU environments expose GPU resources to tenants; classify by underlying temporal, spatial or API mechanism.	Context or remoting subtype
API interception/remoting	GPU calls are intercepted or forwarded to a local/remote GPU service; record transport and whether the target is one node/device.	Core API family
Custom GPU schedulers	Policies choose allocation or execution order; record the underlying sharing primitive.	Context or hybrid
Hybrid approaches	Combine two or more primitives, such as partitions with time-slicing.	Cross-cutting category

The motivation is a possible trade-off between utilization/tenant density and predictable per-tenant performance/isolation. It is a review hypothesis, not a result: some workload mixes may benefit from partitioning or sharing without the expected penalty, while others may show

contention or stranded capacity. The relevant operating-systems concerns include time-sharing versus partitioning, throughput versus fairness, utilization versus isolation, and performance versus predictability. Contention can occur for compute, memory capacity, bandwidth, cache and scheduling opportunities; the causal mechanism and magnitude must be supported by individual studies rather than assumed.

Inference often has request-response behavior, so average latency may conceal slow requests. A reported p95 or p99 is the corresponding empirical latency quantile under a specified load, observation window and measurement method; it does not by itself prove a service-level objective (SLO) was met. Training-only behavior cannot be substituted for inference-serving evidence. GPU sharing should therefore be appraised under concurrent tenants, with workload, request rate, batching and baseline made explicit.

Table 2: Evaluation aspects and operational interpretation

Aspect	Purpose and evidence to record
GPU utilization	Instrument, sampling interval, denominator and activity definition; do not equate device activity with useful work.
Tenant density	Number of concurrent eligible workloads per physical GPU at a stated performance threshold and resource allocation.
Throughput	Completed inference requests or tokens per time unit, with batching and input/output lengths when relevant.
Latency	Defined response-time interval and mean/median statistic at a stated offered load.
Tail latency	p95/p99 or other reported quantile, sample size/window and SLO condition.
Predictability	Within-tenant variability or ability to sustain stated latency/throughput targets across co-tenant mixes.
Resource isolation	Separate compute interference, memory/bandwidth partitioning and any security claim; record measurement versus specification.
Fault isolation	Whether a fault in one tenant is contained from others, with the actual fault model and evidence.

This topic is relevant to inference-as-a-service, multi-model serving and edge systems. Existing studies can differ in hardware, model, batch size, baselines and metrics, so a percentage from one study cannot automatically be compared to a number from another. Examples identified for search-strategy validation include PipeSwitch (temporal switching), REEF (preemptive inference scheduling), gpulets (spatio-temporal partitioning), and ParvaGPU (MIG/MPS spatial sharing) [4, 5, 6, 7]. Their publication identities were checked against conference or publisher records; they are *candidate seeds*, not studies selected by this review. A prior survey covers the broader multi-tenant inference stack and informs an overlap check, but is not a primary study [8]. The protocol-led method follows software-engineering review guidance [9]; PRISMA 2020 informs transparent selection reporting [10].

2 Research Objective

The main objective is to systematically identify, classify and synthesize peer-reviewed primary evidence on sharing one physical GPU among concurrent machine-learning inference tenants. The core comparison concerns time-slicing, NVIDIA Multi-Process Service (MPS), Multi-Instance GPU (MIG) and GPU API interception/remoting. GPU virtualization, custom scheduling and hybrids are coded as implementation layers or combinations when their inference evidence can be attributed to a core family.

The review will examine GPU utilization, tenant density, inference throughput, average/median and tail latency, predictability, cross-tenant interference, resource isolation and fault isolation. It will record the conditions in which each result was obtained, assess methodological strengths and limitations, and develop conditional guidance for engineers. A weak or absent comparison will be reported as a gap, not filled by presumed performance. The final primary-study target in the course guideline is approximately 15–20; this is a planning target, never a reason to relax eligibility or invent studies.

3 Scope and Boundaries

The unit of interest is sharing a single physical GPU for multiple inference tenants, even if the wider serving system uses several nodes or the GPU is accessed through an API proxy. A cluster paper can qualify only if separately reported same-GPU inference-sharing measurements can be extracted. API remoting is eligible only when the remoted endpoint is a shared GPU and its tenant-level effects are separable; mere remote execution is insufficient. Mixed training/inference work can qualify only if the inference subset is separable. GPU model serving, deep-learning inference, and related request-serving workloads qualify; training-only, non-GPU accelerators, pure CPU scheduling and cluster-only placement do not.

Eligibility is *conjunctive*: a paper must meet the publication, mechanism, concurrent-inference, single-GPU evidence, evaluability and date requirements in Section 7. A study does not qualify merely because it mentions one keyword. Surveys, position papers, vendor manuals and preprints may help vocabulary, references and contextual explanation, but are logged outside the primary-study set. A preprint with a later peer-reviewed version is resolved to the final version.

4 Research Questions

Five questions cover mechanism classification, measurement, outcomes, isolation and conditional choice. The taxonomy is a mapping device within a full evidence review, informed by the systematic mapping distinction of Petersen *et al.* [11].

Table 3: Research questions and required evidence

ID	Research question	Evidence/output
RQ1	What time-slicing, MPS, MIG and API remoting mechanisms are evaluated for concurrent inference on one GPU, and how do sharing granularity, layer and isolation boundary classify them?	Mechanism taxonomy and design/evidence table.
RQ2	How do studies measure utilization, throughput, density, average/tail latency, predictability and interference, and how comparable are their experimental methods?	Metric definitions, hardware/workload/baseline matrix and comparability flags.
RQ3	Under what GPU, workload, batch, tenant mix and configuration conditions are changes in utilization, throughput, density and latency reported?	Conditional outcome tables, direct contrasts and uncertainty.
RQ4	What resource and fault isolation properties, trade-offs, implementation costs and conflicting results are reported, and what limits the evidence?	Isolation evidence matrix, quality appraisal and methodological gaps.
RQ5	Which approach or combination is supported for specified inference-serving conditions and priorities, with what confidence and alternatives?	Conditional decision matrix; no universal winner presumed.

5 Search Strategy

A predefined, reproducible strategy combines database searching and separately logged citation chasing. Its objective is reasonable coverage of peer-reviewed systems literature rather than an assertion that one index contains every paper. The expressions below are complete *planned* inputs. Database-specific field settings, exact accepted strings, search dates and returned counts belong in the search log when the searches are run. PRISMA-S informs reporting of the full search method [12].

5.1 Selected academic databases

The course guideline prefers at least four suitable sources from NED's E-Databases section. NED's E-Resources and Research Guide pages are the selection basis [13, 14]. Table 4 specifies four formal search interfaces. Institutional listing does not establish current individual entitlement; if one source is unavailable, the attempt and justified replacement must be documented by amendment.

Table 4: Selected search resources and purpose

Database/platform	Role	Rationale and handling
IEEE Xplore	Formal	Engineering, computing, architecture and conference/journal coverage; test metadata fields and export.
ACM Digital Library	Formal	Computing and systems proceedings/journals; record Advanced Search scope and generated query.
ScienceDirect	Formal	Elsevier journal/book-series coverage; log exact field scope and filter behavior, avoiding an unsupported claim of complete conference coverage.
Springer Nature Link	Formal	Springer Nature computing/engineering coverage; document whether search targets metadata or full text.
Scopus, Web of Science	Citation check	Cross-publisher indexing and citation checking if available; if promoted to formal searches, record full runs and amend protocol.
Google Scholar; publisher references	Discovery	Backward/forward citation leads and access routes; not a substitute for formal runs. Record every discovered report's provenance.

5.2 Search keyword framework

The two mandatory concept groups are sharing/mechanism and inference. Group C is *not* mandatory in the main query, because papers can report outcomes without naming them in metadata. The executable inference block uses the broad term **inference** (which also matches phrases such as "DNN inference" when indexed literally) plus explicit serving phrases that need not contain that word. More specific inference phrases in the keyword framework are candidates for documented sensitivity refinements, rather than an assertion that every variation is separately pasted into each expression. Search sensitivity is checked against independently identified candidate publications across the mechanism families; a seed is not an included primary study.

Table 5: Concept groups and candidate terms

Group	Role	Terms and variations
A. GPU sharing	Mandatory	GPU sharing, multi-tenant/multitenant GPU, GPU virtualization, partitioning, isolation, consolidation, scheduling, time-slicing/time sharing; MPS/multi-process service; MIG/multi-instance GPU; API/CUDA remoting, interception and proxying. Pair ambiguous acronyms with GPU.

Group	Role	Terms and variations
B. Inference	Mandatory	Planned core block: inference, model serving, ML serving, LLM serving, DNN serving. Additional variants for pilot/seed checks: deep-learning inference, DNN inference and inference serving (covered by the unquoted **inference** term where indexed).
C. Outcomes	Screen/extract	GPU utilization, throughput, tenant density, latency, p95/p99, SLO, interference, predictability, resource and fault isolation. Not required in core query.

5.3 Search strings by database

Each core search combines sharing/mechanism terms with inference/serving terms using AND. Three core expressions per database cover broad sharing, named temporal/MPS/MIG mechanisms and API remoting. A fourth, narrowly targeted expression per database seeks pre-2020 MPS evidence for the documented older-study exception. Acronyms occur only with GPU context. Execution, any interface adaptation, field settings and actual results must be recorded verbatim in the later search log. Official IEEE, ACM, Elsevier and Springer search help informed the entry choices [15, 16, 17, 18].

IEEE Xplore Command Search. Select *All Metadata*; enter each expression separately. The field label follows IEEE's Command Search conventions. Apply publication years 2020–2026 and record the interface's exact date range, type and language filter settings [15].

```
I1 = ("All Metadata":"GPU sharing" OR "All Metadata":"GPU partitioning"
 OR "All Metadata":"multi-tenant GPU" OR "All Metadata":"GPU virtualization")
 AND ("All Metadata":inference OR "All Metadata":"model serving" OR
 "All Metadata":"ML serving" OR "All Metadata":"LLM serving" OR
 "All Metadata":"DNN serving")

I2 = ("All Metadata":GPU) AND
 ("All Metadata":MPS OR "All Metadata":"multi-process service" OR
  "All Metadata":MIG OR "All Metadata":"multi-instance GPU" OR
  "All Metadata":"time slicing" OR "All Metadata":"time sharing")
 AND ("All Metadata":inference OR "All Metadata":"model serving" OR
 "All Metadata":"ML serving" OR "All Metadata":"LLM serving" OR
 "All Metadata":"DNN serving")

I3 = ("All Metadata":GPU) AND
 ("All Metadata":"API remoting" OR "All Metadata":"CUDA remoting" OR
  "All Metadata":"API interception" OR "All Metadata":"GPU proxy")
 AND ("All Metadata":inference OR "All Metadata":"model serving" OR
 "All Metadata":"ML serving" OR "All Metadata":"LLM serving" OR
 "All Metadata":"DNN serving")

H1 = ("All Metadata":GPU) AND
 ("All Metadata":MPS OR "All Metadata":"multi-process service" OR
  "All Metadata":"multi process service")
 AND ("All Metadata":inference OR "All Metadata":"model serving" OR
 "All Metadata":"DNN serving")
```

ACM Digital Library Advanced Search. Set the collection to *ACM Guide to Computing*

Literature where available to capture records beyond ACM-published full text. Enter each expression separately in the chosen search box and retain its generated query and field scope. Limit years 2020–2026; document language and peer-reviewed document-type filtering where supported [16].

```
A1 = ("GPU sharing" OR "GPU partitioning" OR "multi-tenant GPU"
 OR "GPU virtualization") AND (inference OR "model serving" OR
 "ML serving" OR "LLM serving" OR "DNN serving")

A2 = GPU AND (MPS OR "multi-process service" OR MIG OR
 "multi-instance GPU" OR "time slicing" OR "time sharing")
 AND (inference OR "model serving" OR "ML serving" OR
 "LLM serving" OR "DNN serving")

A3 = GPU AND ("API remoting" OR "CUDA remoting" OR
 "API interception" OR "GPU proxy")
 AND (inference OR "model serving" OR "ML serving" OR
 "LLM serving" OR "DNN serving")

H2 = GPU AND (MPS OR "multi-process service" OR
 "multi process service")
 AND (inference OR "model serving" OR "DNN serving")
```

ScienceDirect Advanced Search. Select *Title, abstract, or author-specified keywords* under *Show more fields*; the general "Find articles with these terms" field searches the body and is not the planned scope. Record the actual selected field and accepted query. Apply the date range and available article-type filters, then verify peer review and eligibility at screening [17].

```
D1 = ("GPU sharing" OR "GPU partitioning" OR "multi-tenant GPU"
 OR "GPU virtualization") AND (inference OR "model serving" OR
 "ML serving" OR "LLM serving" OR "DNN serving")

D2 = GPU AND (MPS OR "multi-process service" OR MIG OR
 "multi-instance GPU" OR "time slicing" OR "time sharing")
 AND (inference OR "model serving" OR "ML serving" OR
 "LLM serving" OR "DNN serving")

D3 = GPU AND ("API remoting" OR "CUDA remoting" OR
 "API interception" OR "GPU proxy")
 AND (inference OR "model serving" OR "ML serving" OR
 "LLM serving" OR "DNN serving")

H3 = GPU AND (MPS OR "multi-process service" OR
 "multi process service")
 AND (inference OR "model serving" OR "DNN serving")
```

Springer Nature Link Advanced Search. Enter each expression in *Keywords*, which Springer states searches title, abstract and body; there is a *Title* field but no documented abstract-only field in its listed options. Pilot the full-text noise and result volume as specified in Section 5.3. If needed, run a supplementary, separately logged Title-field precision search while retaining recall coverage from the Keywords runs. Apply the publication-date range and filter out books/chapters at screening or through a documented interface filter [18].

```
S1 = ("GPU sharing" OR "GPU partitioning" OR "multi-tenant GPU"
 OR "GPU virtualization") AND (inference OR "model serving" OR
 "ML serving" OR "LLM serving" OR "DNN serving")
```

```
S2 = GPU AND (MPS OR "multi-process service" OR MIG OR
 "multi-instance GPU" OR "time slicing" OR "time sharing")
 AND (inference OR "model serving" OR "ML serving" OR
 "LLM serving" OR "DNN serving")

S3 = GPU AND ("API remoting" OR "CUDA remoting" OR
 "API interception" OR "GPU proxy")
 AND (inference OR "model serving" OR "ML serving" OR
 "LLM serving" OR "DNN serving")

H4 = GPU AND (MPS OR "multi-process service" OR
 "multi process service")
 AND (inference OR "model serving" OR "DNN serving")
```

Date and adaptation rule. I1–S3 are the twelve main-window queries (1 January 2020 through the actual 2026 run date); H1–H4 are four supplementary pre-2020 MPS queries (all indexed years through 31 December 2019). Query identifiers are protocol labels, not text to paste. If an interface rejects an expression or cannot set the historical date boundary, log the error and the closest semantically equivalent accepted query/filter, preserving both. Record any platform-specific stemming, phrase interpretation, collection scope, field and export limit rather than assuming them. The final report will print the exact executed strings, fields, years and real results by database and search lane.

For each platform, merge its core and historical-lane exports, then deduplicate across sources. The validation seeds are PipeSwitch, REEF, gpulets and ParvaGPU, whose publication records are independently available [4, 5, 6, 7]. Inspect approximately the first 20 pilot hits per expression for false matches and record whether each seed indexed in that platform is retrieved; a seed absent from a publisher-specific collection is not automatically a query failure. For broad full-text fields, record the proportion of clearly irrelevant pilot hits and the observed result volume; refine a noisy expression by narrower phrases or a logged supplementary Title-field search, while retaining a recall-oriented route for papers whose title omits the mechanism. If an expected indexed paper is missed, revise terms and log the original and revised accepted expressions, reason and coverage effect. Outcome terms remain optional. The *search-run log* records platform, interface, exact accepted query, field, filters, access route, date/time, searcher, returned/exported counts, export filename/hash and notes. Citation chasing receives a separate source, date, direction and count. No numbers are inserted before real runs.

6 Publication Period

The formal main window begins 1 January 2020 and ends on each actual final search date in 2026, which will be recorded. This follows the guideline's recent-window preference and captures currently relevant MIG-era inference work without pretending that all foundational work began in 2020. To reduce historical bias for MPS, a prespecified targeted search (H1–H4) covers each database's indexed records before 1 January 2020, with no arbitrary lower year boundary; backward citation leads are logged separately. The historical lane is a discovery route, not automatic eligibility: an older peer-reviewed study may enter only through an individually logged exception if it introduces or establishes the evaluated mechanism, or supplies a directly relevant inference-sharing baseline that the main-window studies do not adequately document. Record discovery route, full-text evidence, specific need, decision by both reviewers and year. Apply IC1–IC5 and the same quality rubric. Tabulate older exceptions separately in the final study characteristics and sensitivity discussion. Older context-only references are not primary-study counts. An update near final submission should

use the same protocol or a justified amendment. No study after a recorded cutoff is silently added.

7 Inclusion Criteria

All conditions in Table 6 must be met at full text. Inference and mechanism evidence cannot be substituted by general claims in an abstract.

Table 6: Conjunctive inclusion criteria

ID	Criterion
IC1	Final, English, peer-reviewed full conference or journal primary study. Verify venue/version; preprints and technical reports alone are context only.
IC2	Presents, evaluates or empirically measures one of the assigned local sharing families (or a classifiable virtualization/scheduler/hybrid) among distinct concurrent tenants/processes/models/applications on one physical GPU.
IC3	Includes ML/DL inference or model-serving workload. Mixed training/inference studies require separable inference results.
IC4	Reports at least one relevant inspectable utilization, density, throughput, latency/tail/predictability, interference, resource isolation or fault-isolation outcome under the sharing setup, with sufficient method and configuration detail to interpret it.
IC5	Provides an extractable within-one-physical-GPU result, comparison or per-GPU co-tenant measurement, even if the containing server or scheduler uses multiple GPUs; enough full text is available for appraisal and extraction. A dedicated single-GPU-only experiment is not required.
IC6	Falls in the main publication window or satisfies a documented seminal exception.

8 Exclusion Criteria

At full-text assessment, assign one primary decisive reason in the order below, with secondary notes if useful. Duplicates and uncollected full texts are accounted for *before* assessed-report exclusions in PRISMA 2020 [10].

Table 7: Full-text exclusion codes

ID	Criterion
EC1	Non-primary or non-peer-reviewed version: survey, tutorial, editorial, position/vision paper, vendor document, preprint-only, poster or abstract-only item. Relevant reviews stay in a separate context/citation list.
EC2	Training-only or no concurrent inference tenants/evaluable inference-sharing outcome.

ID	Criterion
EC3	Hardware-only or microarchitecture-only work with no evaluated systems-level sharing mechanism.
EC4	Cluster/multiple-GPU scheduling or remote execution reporting only aggregate system results, with no extractable concurrent co-tenant evidence on one physical GPU. A multi-GPU paper is not excluded solely for its scale.
EC5	Non-GPU accelerator without a relevant evaluated GPU mechanism.
EC6	Insufficient full-text methods, workload/configuration or outcomes to answer any RQ.
EC7	Outside date window without accepted seminal exception, or non-English full text.

Identical imports are linked and removed before screening. Multiple reports of one study are linked and the most complete peer-reviewed report is prioritized; distinct results in companion reports remain traceable. For a report that cannot be obtained after documented library, repository and author-contact attempts, log "report not retrieved," not EC6. Record whether a paper is inaccessible or merely unassessed; neither supplies a fabricated result.

9 Screening and Study Selection

Study selection will follow PRISMA 2020 reporting categories [10], retaining separate record, report and study identifiers, independent votes and reconciled counts.

9.1 Seven-step workflow

Step 1: Database searching. Run each frozen string and record interface, exact query, filters, date/time and counts. Save raw exports untouched.

Step 2: Result export. Export title, author, venue, year, abstract, DOI/URL and source-query fields where available. Assign immutable imported record IDs (R001, etc.).

Step 3: Deduplication. First match normalized DOI; then review title, authors and year manually. Keep all source links, identify a canonical record, and record removed-before-screening counts. Do not conflate records with unique studies.

Step 4: Title screening. Two independent reviewers mark include, exclude or uncertain. Keep uncertain records if title alone cannot establish ineligibility.

Step 5: Abstract screening. Apply the same predeclared criteria independently to retained records. Record both initial votes and a short reason; resolve disagreement before full-text retrieval.

Step 6: Full-text screening. Assign report IDs (P001, etc.), log retrieval attempts and PRISMA reports not retrieved. Two reviewers independently apply all criteria to available reports. Record one decisive EC code with explanation and page/section if excluded.

Step 7: Final selection. Link eligible reports to unique study IDs (S001, etc.). Only eligible primary studies proceed to quality assessment, extraction and synthesis.

9.2 Decision register and reviewer agreement

The spreadsheet will store immutable R/P/S identifiers; title, DOI, source database(s), source query/run, year and version; stage and reviewer; each independent include/exclude/uncertain vote; criterion, short justification and timestamp; adjudicator decision and reason; retrieval status and attempts; primary full-text EC code; and version links. A criteria pilot on approximately ten diverse records will identify ambiguous interpretations before large-scale screening. Each record receives two independent votes at title/abstract and full text; the third group member adjudicates unresolved differences. Preserve initial and final decisions rather than overwriting them. Consultation with the supervisor can be logged for unresolved scope issues, not used to invent eligibility.

9.3 PRISMA count reconciliation

From the register compute records identified by each database/other source, duplicate and other pre-screen removals, unique records screened, records excluded, reports sought, reports not retrieved, reports assessed, reports excluded by primary reason, and unique included studies. Check that identified minus pre-screen removals equals screened; screened minus excluded equals reports sought (adjusting explicitly for record-to-report links); sought minus not retrieved equals assessed; assessed minus excluded equals included reports; then map included reports to studies. If a report maps to more than one record or a study has multiple reports, disclose the mapping and reconcile counts. Counts are reported from the later executed search, not estimated in this protocol.

10 Quality Assessment

Ten criteria use a 0–2 scale: score **0** when absent/unclear, **1** when partly addressed and **2** when sufficiently supported. Maximum is **20**. A score is an appraisal of reporting and internal validity relevant to this review; it is not a publication prestige measure or an automatic inclusion gate. For each criterion record a brief rationale and source page/table/figure. Two reviewers score independently and reconcile differences, preserving initial scores.

Table 8: Ten-item study appraisal rubric; requirements for score 2

ID	Criterion	Score 2 requires
Q1	Clarity of objective	The problem, proposed mechanism and evaluated questions are clear.
Q2	Technical description	Sharing primitive, configuration and relevant implementation layer are described sufficiently to classify it.
Q3	Experimental transparency	Setup, measurement procedure, repetitions or analytical method and result provenance are clear.
Q4	Hardware/workload description	GPU model/generation, software environment and concurrent inference workload/tenant characteristics are reported.

ID	Criterion	Score 2 requires
Q5	Baseline fairness	A relevant baseline is specified and comparable workload, resource allocation and measurement conditions are documented or justified.
Q6	Utilization/throughput	Any claimed utilization, throughput or density benefit uses a defined metric, unit and method; a paper without these claims explains its alternate outcome.
Q7	Latency measurement	Claimed latency results specify interval, load, statistic and measurement method; tail/SLO results include p95/p99 or stated quantile when claimed.
Q8	Multi-tenant interference	Multiple concurrent tenants/co-tenant conditions are tested and sufficient evidence supports interference conclusions.
Q9	Isolation evaluation	Any claimed resource, security or fault isolation has a defined boundary and direct test or appropriately delimited evidence.
Q10	Replication/limits	Details/artifacts permit informed replication and threats, limits or configuration dependence are discussed.

For criteria tied to a claim that a paper genuinely does not make, record a reasoned N/A instead of assigning an artificial failure; report the raw sum, applicable maximum and normalized percentage. The team will check whether this rule produces inconsistent scoring during a two-study calibration before freeze and amend it if needed. Eligible lower-quality studies remain in the synthesis. Report whether major conclusions change when studies below 60% of applicable points are excluded in a sensitivity view; these thresholds guide interpretation, not study selection. Do not hide disagreement or aggregate dissimilar evidence mechanically.

11 Data Extraction Fields

Use a structured form consistently for every eligible study. Categories A–M include source anchors and comparator conditions. Use one row per *study* × *mechanism/configuration* × *inference workload/contrast* × *outcome*; maintain a study-level identity sheet to avoid counting rows as independent studies. Enter **NR** (not reported) for absent values, not zero or an estimate. Mark an author's claim separately from a direct measurement. A second reviewer checks headline numerical outcomes against the source PDF.

Table 9: Extraction form categories

Category	Fields
A. Bibliographic information	R/P/S IDs, title, authors, year, venue/type, DOI/publisher URL, peer-review/version state, search provenance, source PDF page/figure/table for each extracted claim.

Category	Fields
B. GPU/system environment	GPU vendor, model/generation, count and memory, OS, driver/CUDA/runtime/framework versions, node/virtualization environment.
C. ML workload	Model/family/size, task, dataset/request traces, input/output characteristics, batch size, arrival rate, warm-up and inference mode.
D. GPU sharing mechanism	Time-slicing, MPS, MIG, API remoting, virtualization, custom scheduler or hybrid; mechanism layer, capability assumptions and implementation version.
E. Scheduling/partitioning	Policy, time slice, MIG profile/number of instances, MPS settings/caps, resource allocation, API intercept/transport and locality.
F. Utilization/throughput	GPU utilization instrument/denominator/interval, completed requests/tokens per second, tenant density with threshold; raw value and units.
G. Latency	Mean/median, p95/p99, variation, SLO/violation rate, clock interval, sample/window, load and exact unit.
H. Multi-tenant interference	Concurrent tenant count and mix, co-tenant load, standalone versus shared baseline, slowdown or interference calculation.
I. Isolation	Performance interference, resource/memory partitioning, security boundary, fault containment; measured/claimed/NR, fault model and evidence anchor.
J. Experimental methodology	Baseline equality, repetitions, error bars/confidence intervals, statistical test, instrumentation, warm-up, threats and reported limitations.
K. Findings/trade-offs	Exact result direction/magnitude and denominator, authors' interpretation, contrary/null results, implementation/operational costs and applicability.
L. Reproducibility	Code, artifact, dataset, trace, setup and access location; availability claim and date checked.
M. Quality assessment	Q1–Q10 initial/reconciled score, applicable maximum, rationales, anchors, reviewer and check status.

Numeric values remain attached to their specific baseline and configuration. Do not convert figures by visual approximation unless the method is stated and uncertainty recorded. If a paper reports both p95 and p99, keep them separate. For hybrid systems, code each underlying primitive plus the hybrid label. For a distributed serving system, extract only separable single-device effects. Preserve verbatim short quotations only when necessary, quoted and page-anchored; the final prose should paraphrase faithfully and cite the source.

12 Evidence Synthesis Strategy

A structured narrative and comparative table synthesis is planned. Studies will likely differ in hardware, workload, co-tenants, baseline and metrics, so a pooled statistical meta-analysis is not presumed. If genuinely compatible contrasts emerge, a later dated amendment must define effect measure, comparability rule and heterogeneity treatment before pooling.

Table 10: Synthesis areas and outputs

Area	Analysis/output
Mechanism comparison	Taxonomy of time-slicing, MPS, MIG, API remoting, contextual virtualization/schedulers and hybrids; map sharing granularity and implementation layer to RQ1.
Utilization/throughput	Compare reported utilization, throughput and tenant density at stated load and baseline; map to RQ2/RQ3.
Latency/predictability	Compare mean/median and p95/p99 with quantile definition, tenant mix and SLO, including negative or null results; map to RQ2/RQ3.
Interference/isolation	Separate tested interference, partitioning, security and fault containment from undocumented vendor/product claims; map to RQ4.
Trade-offs	Trace utilization/density against per-tenant latency and isolation with comparable baseline conditions; map to RQ3–RQ5.
Research gaps	Report missing direct comparisons, weak baselines, unreported workloads/percentiles and limitations; map to RQ2/RQ4/RQ5.

Comparison rule. Separate within-paper head-to-head comparisons from cross-paper observations. Cross-study numbers are only juxtaposed in hardware, model, batch, concurrent-tenant count, load, metric and baseline strata sufficiently similar to interpret; otherwise describe qualitatively and explicitly state non-comparability. Explain conflicting findings with observed configurations and study limitations without speculating beyond evidence. Integrate the quality rubric and repeat key conclusions without low-quality studies to check robustness. A conditional decision matrix will list scenario/priorities, eligible mechanism, observed benefit, sacrifice, deployment constraint, alternative, evidence sources and confidence. It must permit "insufficient evidence." No universal superior mechanism is assumed.

13 Protocol Changes and Governance

Major changes after searching starts must be dated, justified and traceable. This includes changes to databases, search fields/terms, date range, eligibility, RQs, quality rubric, extraction or synthesis. Pilot refinements are also retained so the final query can be reconstructed. Editorial spelling and layout corrections do not constitute methodological changes. Search exports and earlier protocol versions must be preserved; changed eligibility requires re-screening of affected records and updated PRISMA counts.

Table 11: Required amendment-register fields

Field	Entry	Why recorded
Date and protocol version	UTC date/time, author and group decision	Establish order relative to search and selection.
Old and new rule	Literal former and revised wording/query	Permit exact reconstruction.
Rationale	Pilot evidence, feasibility or discovered ambiguity	Explain the methodological decision.
Impact and remedy	Affected runs, records, counts, re-search or re-screen action	Keep study selection and PRISMA flow consistent.

14 Protocol Summary

This protocol defines a systematic review of GPU sharing for multi-tenant inference on one physical GPU. Four selected academic database interfaces will be searched with twelve logged main-window expressions and four targeted older-MPS expressions. The main primary-study period is 2020 through the actual final search in 2026, with individually justified older seminal exceptions. Two reviewers will independently screen and appraise peer-reviewed primary studies, log retrieval and disagreements, and construct a PRISMA 2020 flow. Structured extraction with page-level evidence will support a condition-aware narrative synthesis of utilization, density, latency, interference and isolation. Major changes after searching will be documented and justified.

A Operational Forms to Complete Before and During the Review

The review log will implement these field specifications electronically, with one row per event or decision and immutable identifiers. Values and counts will be entered from actual searches and screening.

A.1 Search-run and seed-recall form

Table 12: Blank search-run field specification

Field	Entry rule
Run ID; operator; UTC/local date/time	Assign once per interface/query/filter combination.
Database; URL; institutional access	Record exact platform, interface version if visible, entitlement or access route.

Field	Entry rule
Query ID; exact accepted query	Preserve literal text copied from interface and original pilot proposal separately.
Field scope; filters	Title/abstract/keywords/full text, date, language, type; note unsupported filters.
Returned and exported counts	Capture real numbers and explain any mismatch.
Export filename/hash; source link	Save untouched RIS/BibTeX/CSV or screenshot, with provenance.
Seed-study recall	List known candidate identifiers tested, retrieved/not retrieved, and why changes were made.
Change and sign-off	Link amendment version and responsible reviewer; do not overwrite earlier run.

A.2 Screening, appraisal and extraction check

At title/abstract and full text, Reviewer A and Reviewer B independently record their initial vote, code and rationale; Reviewer C records any adjudication. The full-text row also records retrieval attempts and the first decisive exclusion reason. A link table maps imported record IDs to reports and unique studies, preserving original source membership. For each included study, one reviewer extracts evidence and a second checks the mechanism classification, baseline and each headline number against a page, table or figure.

Table 13: Blank decision and extraction field specification

Entity	Required traceability
Record R#	Source run, DOI/title, duplicate linkage, two independent votes, reason and final decision.
Report P#	Full-text URL/version, retrieval status and attempts, two votes, exclusion reason and adjudication.
Study S#	Linked eligible reports, mechanism family, publication identity, peer-review verification and seminal-exception note if any.
Appraisal Q1–Q10	Reviewer scores, rationales, anchors, reconciliation, applicable maximum and sensitivity flag.
Outcome row	Study/mechanism/workload/contrast, baseline, cotenants, hardware, exact value/unit/statistic, source anchor and independent check.
Interpretation row	RQ, direct or cross-study comparison, supporting and conflicting evidence, limitation and confidence.

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